

KNOW YOUR FIRM: MANAGING SOCIAL MEDIA ENGAGEMENT TO IMPROVE FIRM SALES PERFORMANCE¹

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We examine the impact of firm social media engagement on sales performance, answering “whether,” “what,” and “how” questions. The study uses a quasi-experimental design in a social e-commerce setting, for which propensity score matching and difference-in-differences methods quantify a mean 20.67% sales increase after firm social media adoption. We also find that firms that sell low-involvement products benefit more from social media adoption, compared to those that sell high-involvement products. Further, in terms of how to manage social media engagement, we find that informative content, in general, is effective for sales of high-involvement products, whereas promotional content, a new type of content discovered in this study, is more beneficial for sales of low-involvement products. Meanwhile, more social media followers generate better firm sales performance. We used instrumental variables and the control function method to address endogeneity issues and conducted robustness checks to support our conclusions. This study sheds light on the value of firm social media, particularly regarding industry differences and firm know-how.

Keywords: Firm social media, sales performance, product involvement, content analysis, quasi-experiment, propensity score matching, difference-in-differences, instrumental variable, control function

Introduction

Social media has become an important online channel for firms to proactively promote brands and products and interactively communicate with consumers (Kaplan & Haenlein, 2010). Firms often create fan pages on social media platforms and generate content to influence the consumer decision-making process.

We propose three research questions related to firms’ adoption and use of social media. First, to what extent does the adoption of social media generate business value, particularly regarding firm sales performance? Second, to what extent, if any, are

there differences in the sales performance of social media among different types of firms, and what causes these differences? Third, based on their nature, how should firms effectively use social media to achieve better sales performance? Answering these questions is important for both researchers and practitioners.

First, although the use of social media involves connecting and engaging consumers, the ultimate goal of firms is to convert consumer interest into firm sales—in other words, social media ROI is very important for firms. Nevertheless, a survey of 427 marketers from U.S. companies found that 80% of companies were unable to quantify the value of their social media efforts (John et al., 2017). At the same time, research on the sales

¹ Bin Gu was the accepting senior editor for this paper. Ali Tafti served as the associate editor.

performance of the firm use of social media (i.e., firm-generated content, or FGC) is limited in comparison with that on consumer behavior. A few extant studies (e.g., Goh et al., 2013; Chen et al., 2015; Kumar et al., 2016) have generated implications for the performance impact of FGC. Nevertheless, we have a limited understanding of industry (or firm) heterogeneity in terms of its influence on the value of FGC, as most of the studies have focused on single firms or industries. While two studies (Chung et al., 2020; Colicev et al., 2016) are based on multiple industries, their analyses are overall examinations without specific investigation into the industry difference.

Second, based on the nature of the industry (or firm), its effect on the consumer decision-making process differs, as does the effectiveness of firm social media content. Acknowledging industry differences in terms of social media impact is important for firms to be able to manage social media efforts. Likewise, social media is particularly suitable for some industries as a means of creating consumer trust² because, when facing different products, the consumer decision-making process can differ at different stages, and having to deal with uncertainty could lengthen the process (Engel et al., 1995). Therefore, the effectiveness of firm social media content depends on the extent to which it addresses consumers' uncertainty and creates consumer trust. Research has not examined industry differences in sales performance as related to a firm's social media content strategy. In this study, we use product involvement as an angle to investigate the performance difference. Since different levels of product involvement involve different consumer behaviors in the decision-making process (Zaichkowsky, 1985), we believe that this would cause different outcomes for firm social media content.

Third, real-world cases have indicated the importance of know-how—that is, knowing how to effectively use social media to achieve better performance. Firms, even in the same industry, could have very different outcomes from their social media efforts. Starbucks is often mentioned as an example of excelling in social media. From campaign content to visual expression, Starbucks masters content that is effective in engaging consumers and therefore generates extremely high social media ROI.³ In contrast, JD Wetherspoon, also in the food chain industry, has not been able to determine an effective way to generate social media content despite their best efforts, resulting in the closure of all of its social media accounts.⁴ In the literature, studies have addressed the “how” question mainly from three aspects, i.e., volume of FGC, communication mode, and FGC content analysis. Given the rich firm social media content available, we believe content analysis could provide

nuanced implications; however, such research is very limited in the literature, let alone research investigating content strategy with respect to industry (or firm) differences.⁵ In this study, we focus on content analysis to investigate how to effectively manage firm social media efforts to improve sales performance according to the firm's nature.

We aim to answer the three research questions proposed above. The launch of Weitao (an integrated microblogging function) on Taobao (China's leading e-commerce platform) in November 2013 provides the means for a quasi-experimental design to quantify the sales effect of firm (Taobao store) social media adoption and engagement. Weitao is the sole social media marketing function embedded in the Taobao platform; when a firm chooses to use Weitao, the firm opens its own microblogging account to promote its store and products. This social e-commerce setting allows us to observe both firm social media engagement and joint firm sales.

Thus, in this study, we examine the firm-level sales performance of firm social media engagement. We examine multiple firms in different industries. We take product involvement as the theoretical perspective to understand the industry difference and to examine how the sales effect of a firm's social media engagement may be different for firms with different levels of product involvement. We also explore how firms manage their social media efforts, especially the content of blog postings, to achieve better firm sales performance and examine how firms manage such efforts based on the nature of their industry. To the best of our knowledge, no prior research has adopted this focus.

We assembled a novel data set from taobao.com and its integrated social media function, Weitao, to examine the sales effect of the firm (Taobao stores) use of social media (Weitao). To address the endogeneity issue caused by firm self-selection (adopting Weitao), we employed a propensity score matching (PSM) method, and used a difference-in-differences (DID) method to quantify the sales effect. Overall, we found that the use of Weitao significantly boosted firm sales. In general, the firms in our study had a mean of 20.67% increase in sales as a result of Weitao adoption.

With respect to industry differences, our results show that firms in low-involvement product industries (stationery goods in this study) benefit more than those in the high-involvement product industries (cosmetics products in this study) from the adoption of Weitao. Specifically, we found that Weitao adoption yielded a mean sales increase of 27.79% for stores that sold stationery

² <https://www.socialmediatoday.com/social-networks/anthonycarranza/2015-06-16/7-industries-benefit-most-social-media>

³ <https://ipa.co.uk/knowledge/case-studies/starbucks-and-social-how-starbucks-created-the-fourth-place-in-coffee-culture>

⁴ <https://www.bbc.com/news/business-43781281>

⁵ Only Goh et al. (2013) had FGC content analysis.

products, whereas stores that sold cosmetics products realized an average sales increase of 21.29%.

Regarding how social media efforts should be managed, we found that when stores made an effort to attract more Weitao followers, they benefitted more in terms of firm sales. However, effort, in terms of more Weitao blog postings, did not generate more sales. We further explored Weitao blog content and found that, overall, informative content was more significant in helping firms (especially those selling high-involvement products) address consumer purchase uncertainty and therefore promoted firm sales. Meanwhile, content analysis suggests a new type of content, i.e., promotional content, which we found to be effective for firms selling low-involvement products.

We used a control function model to address the hidden bias issue in the PSM, as well as instrumental variables to address endogeneity and a first differencing model to tease out firm heterogeneity. Our results are robust to various tests.

This study contributes to the value of social media FGC literature in the following ways. First, it contributes to the limited research on the firm performance of FGC by explicitly examining industry differences. The unique research setting allowed for a more accurate quantification of the sales effect of firm social media efforts, which thereby answered the “whether” question. Second, this study provides an angle, i.e., product involvement, to understand how industry differences in sales performance relate to firm social media, thereby answering the “what” question. Product involvement, in terms of the consumer decision-making process, which can be influenced by social media content, can shed light on what causes the different effects of social media content. Third, the study enriches the understanding of the effectiveness of firm social media efforts, answering the “how” question. By analyzing different aspects of firm efforts, particularly through content analysis, we improve the understanding of how to manage FGC according to the nature of the firm and industry.

Literature Review

The Value of Social Media FGC

Social media provides two-way communication for firms to proactively market products and connect with potential customers (Kim & Ko, 2012). This study concerns the value of social media FGC, which has two substreams in the literature, i.e., the impact of social media FGC on consumer behavior and the impact of social media FGC on firm performance. We provide a summary of the two substreams with the key information of each study in Appendix A.

Impact on Consumer Behavior

The first substream of research concerns the impact of social media FGC on consumer behavior. The extant research in this stream focuses on the consumer behavior side by investigating the behavioral outcomes of FGC or on the FGC side by investigating the content. For the former substream of research, studies overall find a positive impact of FGC on consumer behaviors such as brand loyalty (e.g., Erdoğan & Cicek, 2012; Rishika et al., 2013), brand trust (e.g., Laroche et al., 2013), customer relationships (e.g., Clark & Melancon, 2013; Gu & Ye, 2014; Kim & Ko, 2012), and consumer word of mouth (e.g., Homburg et al., 2015; Meire et al., 2019; Song et al., 2019). In these studies, FGC is often treated as an indicator of *whether a firm opens a fan page or whether consumers participate in a firm social media community*, and surveys are the dominant method used to measure consumer behavior. To fully understand the impact of FGC, however, we also need to examine whether the consumer behavior elicited by FGC translates into better firm performance in terms of sales. However, very few studies have examined this relationship. Two exceptions are John et al. (2017) and Song et al. (2019). While John et al. (2017) investigated the causal link between consumer behavior (“liking” a brand page) and consumer purchasing, their study focuses on the later stage of the link (i.e., from consumer behavior to firm performance) but not on the first stage (i.e., from FGC to consumer behavior), and firm performance is measured mainly by consumers’ propensity to purchase, based on lab experiment data. Song et al. (2019) found that movie FGC can predict box office revenue indirectly via user-generated content (UGC).

Another focus of the first substream of research on the FGC side has investigated content (or the frequency of content) and its impact on consumer behavior. In such research, consumer behavior, such as following, liking, or commenting, has been treated as a token act (e.g., Lee et al., 2018; Meire et al., 2019; Miller & Tucker, 2013; Colicev et al., 2018). These studies have found that active social media engagement and informative content can stimulate positive consumer behavior. However, as Chung et al. (2020) note, although the measurement of content in these studies is appropriate for individual-level analysis (for consumer behavior), it may not be applicable to an examination of the impact of FGC content strategy at the firm level. In the social e-commerce context of our study, the objective of FGC may be directed more toward sales performance than toward eliciting consumer behavior in terms of “liking” or commenting. Further, how firms should manage content according to the nature of their firm to generate better firm performance is also poorly understood. Thus, we investigated the impact of firm social media efforts on sales performance and conducted content analysis to examine the sales effect of FGC content types for different types of firms.

Impact on Firm Performance

The second, much smaller substream of research concerns the impact of social media FGC on firm performance. Nevertheless, the studies in this substream greatly contribute to the understanding of the firm-level value of FGC and to the objective measurement of firm performance. The overall conclusion of these studies is that social media FGC has a positive influence on firm performance in terms of sales (e.g., Goh et al., 2013; Chen et al., 2015; Kumar et al., 2016), stock market performance (e.g., Chung et al., 2020), and brand value (Colicev et al., 2016).

Since this substream of research closely relates to our study, we discuss it in depth, focusing on three areas: single-industry (or single-firm) studies, multiple-industry studies, and the management of FGC. Most studies in this substream are based on data from a single industry or firm, e.g., an apparel firm in Goh et al. (2013), a wine-selling specialty firm in Kumar et al. (2016), the music industry in Chen et al. (2015), and the financial services industry in Hewett et al. (2016) (see the complete list of studies in Appendix A).

There are two studies in this substream of research that examine multiple firms across different industries. Chung et al. (2020) studied 63 firms across 13 industries (e.g., financial services, information and communication, durable manufacturing) and compared their financial returns (measured by abnormal returns and Tobin's q) to firm communication actions on social media. Based on a data set of 87 brands across 17 industries (e.g., financial services, technology, automotive, luxury), Colicev et al. (2016) examined the influence of brand actions on brand value. The results of these studies are more generalized than those of single-industry studies. However, neither investigates industry heterogeneity.

The third focus of this substream of research is how to manage FGC to influence firm performance. Extant studies have addressed this issue mainly according to three aspects—the volume of FGC, communication mode, and FGC content analysis. For the volume of FGC, studies have consistently found that the volume of FGC has a positive effect on firm performance (e.g., Song et al., 2019; Chung et al., 2020; Hewett et al., 2016; Colicev et al., 2016. See Appendix A for details of each study). Regarding the communication mode, studies have identified effective ways for firms to communicate with customers on social media. Goh et al. (2013) found that the direct communication mode is more effective. Chen et al. (2015) showed that for music sales, broadcasting personal messages on social media is much more effective than automated messages. In the context of a media company's microblogging, Gong et al. (2017) determined that the firm's direct tweets and influential retweets both had a

positive effect on the viewing of the company's shows, but in different ways. Chung et al. (2020) found that the timely response to negative customer messages is associated with increased firm market performance. Regarding FGC content analysis, only Goh et al. (2013) addressed this issue, finding that the effect of FGC is related to persuasive content, measured by valence. We believe that content analysis could provide many insights, given the availability of rich firm social media content. Therefore, in this study, we conducted content analysis to answer the "how" question and investigated firms' effective content strategy with respect to the nature of the industry (or firm).

Overall, the limited studies in this substream of research have specifically examined neither the impact of FGC on firm performance in terms of industry differences nor how to manage FGC according to the nature of the industry. Due to the nature of the industry (or firm), social media content for different types of products may be perceived differently by consumers and may therefore influence consumers differently. In this study, we examine the sales impact of FGC and the industry difference, taking product involvement as a means to differentiate industries and gain insights into industry nature. We also investigate how FGC can be effectively managed with a focus on content analysis.

Product Involvement

Product involvement is commonly defined as the consumer's enduring perceptions of the importance of the product category, based on the consumer's inherent needs, values, and interests (Zaichkowsky, 1985). Product involvement is a causal or motivating variable with a number of consequences for consumers' purchase behavior and communication patterns. Taking consumers as a group, products differ in their tendency to motivate involvement. Low-involvement products refer to products that most consumers perceive as having little relevance to their core values and the product class and are associated with low brand commitment, whereas high-involvement products refer to products of high value to consumers or to products that represent a high level of purchase risk for consumers (Lastovicka, 1979). Typically, low-involvement products are associated with a low risk of purchase failure or have a low price tag, whereas high-involvement products have a high price tag (Assael, 1984).

Studies have shown that product involvement influences the consumer decision-making process in terms of consumers' willingness to search for information about the product, their perception of alternatives in the same product category, and brand loyalty (Belk, 1982; Quester & Lin, 2003; Vann, 1984; Zaichkowsky, 1985). Consumers typically go through different decision-making processes based on product

involvement (Assael, 1984; Engel et al., 1995). For instance, the purchase of high-involvement products involves more information searching due to the high value or high-risk nature of the product (Zaichkowsky, 1985). Based on the literature, we believe that product involvement is a useful classification for understanding the effect of firm social media engagement, as social media content may influence certain stages of the consumer decision-making process pertinent to product involvement.

Researchers have developed product involvement measurements (Bloch, 1981; Zaichkowsky, 1985) such as the Personal Involvement Inventory (PII; Zaichkowsky, 1985). Based on the PII, cosmetics products are high involvement (Brisoux & Cheron, 1990) because they directly relate to consumer self-image and have many attributes that make it difficult to decide which item to purchase (Park & Moon, 2003; Suh & Youjae, 2006), whereas stationery goods (such as pens) are low involvement (Mittal, 1989) because they are consumable and considerable involvement is not needed for their purchase (Martin, 1998). In this study, we follow these identifications.

Prior literature (e.g., Gu et al., 2012) has examined the influence of product involvement on the effect of word of mouth (WOM). Gu et al. (2012) found that for firms selling low-involvement products, internal WOM has a greater influence on sales than external WOM, whereas for those selling high-involvement products, internal WOM has limited influence and external WOM has a significant impact on sales. However, the role of product involvement in the context of social media has not been examined.

In this study, by interactively providing product and brand information, we take the product involvement perspective to discern the extent to which firm social media engagement influences the consumer purchase decision for products of different levels of involvement. Following the literature, we chose cosmetics products as the high-involvement product and stationery goods as the low-involvement product.

Hypothesis Development

We develop hypotheses for the relationship between firm social media engagement and sales performance based on consumer decision process theory (Engel et al., 1995) and social presence theory (Short et al., 1976). The consumer decision-making process refers to consumers' process when purchasing a product and can be divided into five stages: need recognition, information searching, evaluation of alternatives, purchase decision, and postpurchase behavior (Engel et al., 1995). By actively sharing product information and

interacting with consumers, firm social media engagement can create social presence, which may affect one or more stages of the consumer decision-making process (Short et al., 1976) and may motivate consumers to make purchases and thus improve sales performance (O'Boyle, 2014).

Firm Social Media Adoption and Sales Performance

In the consumer decision-making process, firm social media may act as an important source of information, which is likely more influential in the stages of information searching, the evaluation of alternatives, and postpurchase behavior. The information richness and interactivity of social media can create social presence, while the nature of the social network reinforces the influence. Overall, firm social media may increase the likelihood that firms' products will be included in consumers' consideration set, thereby impacting consumers' purchase decisions.

Information searching is a critical stage in the decision-making process of most consumers (e.g., Assael, 1984; Punj & Staelin, 1983). Once they have identified a need, consumers start to search for the right product. Content generated by firms is an important information source for consumer searching, as it serves as the single authoritative source of product and brand information (Cole et al., 2011). Firm social media provides a significant amount of content about the brand and its products and may include product functions and qualities, new releases, and promotions. Firm social media also shares trend and product use tips that attract and educate consumers. In social media, such information can reach a larger audience and spread faster due to the network effect (Kwak et al., 2010). Furthermore, social media has the capability to create social presence (based on the sociability, interactivity, and personalization it facilitates), which can transmit rich information (Fulk et al., 1987). Therefore, at the stage of information searching, social media FGC can attract more consumers by providing rich information, eliciting their interest, and stimulating demand.

The *evaluation of alternatives* stage is when consumers decide which product attribute is important (Engel et al., 1995). In the evaluation process, consumers are looking for tangible and reliable information that will allow them to make a choice. Firm social media content, especially concrete content about product features, adds value to the evaluation process and can differentiate the focal firm from its competitors. In addition, the interactive feature of social media can effectively address consumers' purchase concerns. Together, these features can enhance consumers' product recognition and reduce uncertainties in the consumer decision-making process (Punj & Staelin, 1983).

When consumers evaluate alternatives, they also consider products from brands with which they have intimate relationships. Social media FGC can foster such intimate relationships by creating social presence (Kim et al. 2020). Specifically, firm social media may act as a social actor by posting “socially rich” content on the brand community page and communicating with consumers. Interactions between the firm and consumers can engender intimacy and immediacy, thereby forming social presence (Han et al., 2015). High levels of social presence can cause consumers to perceive a feeling of human contact, sociability, warmth, and sensitivity (Short et al., 1976), which contributes to positive consumer attitudes and influences aspects such as brand attitude, store image, and purchase intentions (Cyr et al., 2007; Moon et al., 2013).

In the *postpurchase* stage, firm social media can influence consumer behavior by facilitating the sharing of shopping and product use experiences, positive or negative, which can impact other consumers’ purchase intentions. Also, the social presence elicited by FGC can generate brand recall and keep the brand at the top of consumers’ minds, which has been shown to improve brand loyalty and lead to repurchase (Kumar & Benbasat, 2002; Lu et al., 2016).

Based on the above discussion, firm social media can improve sales performance by influencing the consumer decision-making process. Accordingly, we propose the following:

H1: *Firm social media adoption has a positive impact on sales performance.*

Industry Differences in the Sales Effect of Firm Social Media Adoption: The Role of Product Involvement

Social media has become a significant channel for firms to influence consumer purchase decisions. This effect, however, depends on the nature of the industry. Product involvement is a critical element that shapes the way consumers search for, process, and save information (e.g., Bloch et al., 1986; Zaichkowsky, 1985) and may result in different firm social media engagement outcomes. Below, we discuss how product involvement causes different types of consumer information search and information processing in the decision-making process and how firm social media can influence the process.

First, consumer information search can be classified into two categories, i.e., ongoing and prepurchase (Bloch et al., 1986). For both types of information search, consumer behavior is subject to the type of product, specifically product involvement (Bloch et al., 1986), and firm social media serves as an important source of information. Ongoing search is

defined as information-seeking activities that are independent of specific purchase needs or decisions (Bloch et al., 1986). Researchers have found that for an ongoing search, product involvement is an important prerequisite and may have a significant influence on consumer purchase behavior (Bloch et al., 1986). When looking for high-involvement products, consumers need more information to eliminate uncertainties and make decisions (Zaichkowsky, 1985). Therefore, they have a high motivation to participate in ongoing search activities, including following firm social media. If managed well, firm social media content can help create product awareness, facilitate interactions with consumers, and stimulate demand. Because ongoing search has no specific target, consumers may browse social media content broadly to obtain a good understanding of products and the brand, leading to a higher likelihood of purchase. For low-involvement products, consumers need less information and therefore have a low motivation to follow firm social media content. The ongoing search is easier and faster to complete, meaning that firm social media likely has less impact on the sales of low-involvement products.

Different from ongoing search that has no specific target, prepurchase search refers to search activities that facilitate specific purchase decision-making (Bloch et al., 1986). Such searches are intermittent but require focused information. Firm social media content can provide detailed product information, and the social media feature can address consumers’ specific queries, enhancing consumer purchase intentions. Prepurchase search is also influenced by product involvement (Bloch et al., 1986). Involvement theory suggests that product involvement can influence the degree of information searching and processing for the product (Belk, 1982; Quester & Lin, 2003; Vann, 1984; Zaichkowsky, 1985). Typically, consumers request more information when purchasing high-involvement products.

Second, for information processing, the effect of firm social media is also subject to the level of product involvement. According to the elaboration likelihood model (Petty & Cacioppo, 1986), the effectiveness of persuasion depends on information processing, for which there are two major routes: central and peripheral. Under the central route, when consumers have a high level of motivation and ability, they are willing to engage in considerable cognitive processing effort, called high elaboration likelihood, to evaluate product-related information. If the product and firm-related information are convincing, consumers will develop positive attitudes and purchase intentions toward the product and brand. Under the peripheral route, however, consumers are either unwilling or unable to exert effort to process information and instead focus on peripheral cues, such as brands or visuals.

In a high-involvement situation, consumers are highly motivated to process social media content information and are thus likely to take the central route. Consumers tend to obtain more information from multiple sources to compare different products for their purchase decisions (Gu et al., 2012). In a high-involvement situation, they are more likely to request more information through social media to see detailed content and will make their decision after scrutinizing the received content. Firm social media thus affects consumer behavior through cognition, thereby influencing the purchase decision.

In contrast, in a low-involvement situation, consumers have little motivation to process firm social media content. Therefore, consumers tend to process information under the peripheral route. They are less likely to request more information and more likely to focus on peripheral information, such as product appearance or the brand itself, to make purchase decisions. Therefore, in this case, firm social media has less influence on consumers' purchase decisions. Based on the above, we propose:

H2: *Social media adoption has a greater impact on sales performance for firms that sell high-involvement products.*

Sales Effect of Firm Social Media Efforts

On social media platforms, firms make efforts to communicate with consumers. At its core, FGC is a process of computer-mediated communications in which interactions between the sender and receiver can change the receiver's behavior or attitudes (Merton & Merton, 1968). Thus, we develop hypotheses for the relationships between firm social media efforts and sales performance based on communication theory (Krone et al., 1987; Mohr & Nevin, 1990). According to communication theory, the sender (firm), receiver (consumer), content, and effect are the main components of communication. Aspects of communication, including extensiveness, intensiveness, and content, all influence communication effectiveness (Mohr & Nevin, 1990). Therefore, firm social media efforts associated with these aspects may impact sales.

Extensive and intensive social media FGC help firms improve consumer awareness (Lou & Yuan, 2019; Hoyer & Brown, 1990), which is an initial step in the consumer decision-making process and a prerequisite for reaching subsequent stages (Lavidge & Steiner, 1961). In the context of social media, the extensiveness and intensiveness of communications are measured by the number of followers and the number of blog postings, respectively (Starbird & Palen, 2012; Mohr & Nevin, 1990).

Firms create content on social media platforms to attract consumers to follow their accounts. In social media, firms act as an information-source type of user that operates as the hub of a social network and usually has a large number of followers (Java et al., 2007). According to information diffusion, hubs are central to information dissemination in any network (Balcan et al., 2009), and information distributed from hubs can spread to a broader population (Starbird & Palen, 2012). Followers can then spread FGC to their friends, creating a ripple effect. The higher the degree of the hub (i.e., the number of followers connected to the hub), the greater the consumer awareness generated by firm social media. With greater consumer awareness, the chance that consumers will buy from the firm is higher. In addition, firm social media not only disseminates information but also interacts with consumers, e.g., by answering their questions, thus increasing the likelihood of having more loyal consumers. This leads to a stronger sales impact when considering loyal consumers' contributions and their recommendations to other consumers on social media (Gremier & Brown, 1999). We therefore propose:

H3: *Firm social media efforts with more followers are associated with better sales performance.*

When consumers follow a firm's social media page, they are continuously exposed to FGC. The volume of content is very important in such a channel because it transmits information, fosters decision-making, and encourages commitment and loyalty (Mohr & Nevin, 1990). Frequent firm communication with consumers can increase consumer awareness and potentially lead to improvements in firm performance. Furthermore, under continuous exposure to FGC, consumers' attitudes toward a brand can be positively changed (Grimes & Kitchen, 2007; Rahman & Soesilo, 2018; Janiszewski, 1993), leading to the mere exposure effect (Zajonc, 1968). The mere exposure effect suggests that the mere repeated exposure to a stimulus object (even if unfamiliar) can enhance an individual's attitude toward the object (Zajonc, 1968). On social media, the number of blog postings captures the intensity of communications between the firm and consumers and has a positive relationship with consumer awareness and product sales (Rui et al., 2013). An effective social media effort may involve more blog postings to maintain regular consumer interaction, generate the mere exposure effect, and strengthen the bond between the brand and consumers. Overall, consumers are more likely to make purchase decisions when they have a closer relationship with the firm (Rishika et al., 2013). Therefore, we propose:

H4: *Firm social media efforts with more blog postings are associated with better sales performance.*

While firm efforts in terms of the number of followers and blog postings affect consumer awareness, firm social media content serves as an important source of information for consumers and can influence consumer need recognition and product evaluation (Vakratsas & Ambler, 1999). Prior studies have noted that different content categories have different impacts on consumer behavior (e.g., Kumar et al., 2016; Rishika et al., 2013). In this study, we analyze the impact of FGC content types on sales performance.

Specifically, content can be classified as primarily informative or persuasive (e.g., Abernethy & Franke, 1996; Bertrand et al., 2010; Liaukonyte et al., 2015). Informative content refers to factual data on the nature and function of a product or service and can reduce uncertainty (Abernethy & Franke, 1996), whereas persuasive content assumes that consumers already have an understanding of the nature and function of the product but need to be convinced of the benefits that distinguish it from alternatives in the market (Armstrong, 2010; Berger & Milkman, 2012). In this study, we analyze which type of content is more effective for sales on social media platforms.

In their decision-making process, consumers are looking for tangible and reliable information. However, markets are characterized by imperfect information, which may deter consumers from learning about a product's existence, quality, and characteristics. Informative content on social media platforms alleviates information deficiency, as consumers can receive detailed and additional information (e.g., origins, performance) at no cost. Informative content can influence multiple stages of the consumer decision-making process by enhancing consumers' product recognition, facilitating information searching and the evaluation of alternatives, and thus increasing purchase intentions. For instance, prior studies have found that providing informative content leads to more immediate online brand searches, while persuasive content has no effect (Chandrasekaran et al., 2018; Du et al., 2019). In addition, informative content helps to reduce uncertainty (Abernethy & Franke, 1996), as it provides facts or cues about specific aspects of a product and is objective rather than subjective (Resnik & Stern, 1977). Furthermore, brand informative content on social media could also have a positive effect on consumer brand attitude, especially in the high-involvement scenario (Lou et al., 2019).

In contrast, persuasive content seeks to induce needs and create product differentiation, influencing the consumer evaluation of alternatives in the decision-making process. The processing of persuasive content is a subjective consumer-pertinent process, whereas that of informative content is an objective product-pertinent process. It is relatively difficult to alter consumers' tastes and persuade them to buy. Furthermore, compared to consumer-initiated persuasive content, firm-initiated persuasive

content is less convincing, especially in the e-commerce context we study. Therefore, we propose:

H5: *Firm informative social media content is associated with better sales performance than persuasive content.*

Data and Model Specifications

Data in this study are obtained from Taobao (www.taobao.com). Taobao has the most extensive collection of products and services and the highest gross merchandise volume of any platform, with about 10 million sellers. Sellers on Taobao open and operate their own stores and market their products on social media platforms to improve performance. However, since one of the most popular third-party social media platforms blocked Taobao's external link, creating fierce competition (Erisman, 2017), Taobao launched Weitao, its own seller-oriented microblogging social media function, on its e-commerce platform in November 2013. Taobao stores can choose to open a Weitao account to market their stores and products. In this case, Taobao stores are attached to a Weitao account that can be used to post microblogs to connect with consumers, which thereby creates their own brand community. Consumers that follow Weitao accounts receive corresponding store updates. However, Weitao is merely the store's social media channel—all transactions are still conducted through the Taobao store.

The launch of Weitao enabled a quasi-experiment to examine the effect of Weitao adoption on store sales performance. We therefore employed a DID method to quantify this effect by examining sales differences from pre- to post-Weitao adoption and between stores with and without Weitao. To deal with the self-selection issue, we used a propensity score matching (PSM) method to match the treatment group (stores with Weitao) with a control group (stores without Weitao) (Rosenbaum & Rubin, 1983). To ensure the validity of the DID approach, we examined differences between these two groups in pretreatment variables and inspected the pretreatment trends of store sales. We further constructed instrument variables to model the store's decision to open a Weitao account and used a control function model to deal with any hidden bias issue due to unobservables in the PSM.

In addition to the overall impact of store social media adoption on sales performance, we examined industry differences by comparing high- and low-involvement industries. We also investigated how to manage social media efforts, with respect to industry nature, to achieve better firm sales performance. The analyses include content analysis for firm-level content types, a first differencing model for the sales effect of social media

efforts (controlling for firm heterogeneity), instrumental variables, and the control function method for endogeneity.

Data

The first step of our data collection procedure was to obtain all stores that opened a Weitaio account. We collected stores in two industries, cosmetics and stationery, with cosmetics representing a high-involvement industry and stationery representing a low-involvement industry, following the literature on product involvement discussed above. We searched the keywords “cosmetics” and “stationery,” respectively, on the Weitaio mobile search page and obtained a list of all Weitaio accounts.⁶ The search results gave us approximately 100 pages, representing all relevant Weitaio accounts at that time. Note that Weitaio was launched in November 2013. In the beginning, only a few stores chose to open a Weitaio account. It took several months until a reasonable number of stores opened Weitaio accounts. By observing this trend, we collected a sample of stores that opened Weitaio accounts between April 2014 and June 2014. We dropped 934 cosmetics and 213 stationery Weitaio accounts from Tmall (B2C) stores and kept only Weitaio accounts from Taobao stores to avoid systematic bias between the two platforms.⁷ For the final Weitaio account list, we retrieved information from each Weitaio account, including the Weitaio account opening date, the store webpage attached to it, all blog postings, and the number of Weitaio followers.

Based on the store webpage information retrieved from the Weitaio account, we obtained the store profile, including the store name, store opening date, store ratings (including store description, service, and logistics ratings), accumulated store credits, the number of consumers who bookmarked the store, the total number of comments that the store received from the store’s opening to six months prior to when the opening of their Weitaio account, and the number of comments that the store received in the previous month and in the previous week. Our final sample consisted of 422 cosmetics stores and 125 stationery stores, i.e., our treatment group, and no store sold both cosmetics products and stationery goods.

To obtain a pool for the control group to use with PSM, we searched the same keywords, “cosmetics” and “stationery,” on the Taobao PC store search page, refined search criteria to Taobao stores only, and recorded all search results. We

dropped all stores that had opened a Weitaio account from our sample, leaving 7,489 cosmetics stores and 4,134 stationery stores. These stores comprised the pool for the control group. We then retrieved the store profile from the store webpage, as we did for the treatment group.

Finally, we needed data for the outcome variable, i.e., store sales. Taobao does not provide real-time sales information on the store’s webpage, it only shows sales for the previous six months (in terms of the number of transactions), the previous month, and the previous week. Because we searched Weitaio for newly opened accounts, data collection at that time point gave us store sales data for the six months (also one month and one week) prior to the opening of the Weitaio account. We waited one week, one month, and six months to collect data again to obtain the respective store sales after they opened their Weitaio accounts. Using this approach, we retrieved sales data for both the treatment group and the pool for control group. We also waited six months to collect the store profile and the information on the number of blog postings, the content of blog postings, and the number of followers from the Weitaio account.

Content Coding Process

We followed a content classification procedure to classify all of the collected Weitaio blog postings, yielding a total of 4,989 blog postings (1,325 from stationery stores and 3,664 from cosmetics stores) in the six months after the Weitaio account opened (note that the platform had a restriction on the maximum number of blog postings a store could post). The procedure included five steps: the orientation period, development of the training set, text mining on the training set, content classification, and factor analysis. We provide a detailed description of the coding procedure in Appendix B.

From the content-coding process, we obtained three factors that we labeled as *informative*, *persuasive*, and *promotional*. Promotional content, which includes promotions and deals, is a new factor (the literature primarily discusses informative and persuasive content). Based on our observation, promotional content is frequently used in Chinese social media marketing, especially in the e-commerce context. We conducted validity checks for the proposed three factors. The Kaiser-Meyer-Olkin (KMO) value was 0.809, indicating a high level of validity. The common variances for all attributes

⁶ Taobao has a tag to differentiate a store’s industry based on the store’s main product category. “Cosmetics” and “stationery” are tags that Taobao uses, which made our search convenient. We dropped stores whose industry did not match our search keywords.

⁷ We dropped Tmall Weitaio accounts based on the following considerations: First, the marketing strategy of Tmall stores is very different from that of Taobao stores. We interviewed a few Tmall sellers and were

told that Tmall stores rely heavily on expensive paid search and display ads as their main marketing tools. Although they opened Weitaio accounts when Weitaio was launched, Tmall stores did not focus on it and their Weitaio accounts were very inactive. As a free marketing tool, Weitaio is mainly intended for Taobao sellers, i.e., the small and medium ones. Second, due to platform data access restrictions, sales information and detailed store information of Tmall stores could not be obtained.

were higher than 0.4—suggesting that the information of all attributes was effectively extracted. We also checked the factor loading coefficients between each attribute and its corresponding factor. All coefficients were higher than 0.4, confirming the validity (Chan & Idris, 2017). The corresponding relationship between content types and attributes is shown in Table B1.

Meanwhile, the two industry subsamples, cosmetics and stationery, showed different patterns of content posting. Specifically, the cosmetics stores posted more informative content—such as information about product origins, quality, performance, and user experience—than stationery stores. Stationery stores posted more promotional content than cosmetics stores and had significantly higher scores for two promotional attributes, namely price and delivery. For persuasive content, we found no significant difference between the two subsamples for either of the two persuasive attributes.

For each blog posting, we calculated the mean value of attributes for each type of content. Then, we calculated the mean value of blog postings at the store level and used them as the values of the store FGC content factors in the sales effect analysis. The store-level content factor scores are statistically summarized in Table 1.

Empirical Specifications

Propensity Score Matching

The objective of our PSM was to match the treatment group with a control group with similar covariate distributions before the store opened a Weitaio account. The process involved two steps. The first step modeled the store's decision to open a Weitaio account as a function of store pretreatment covariates. We used a generalized boosted regression model (GBM) to estimate propensity scores (McCaffrey et al., 2004) because this algorithm can adaptively choose variables to include in the propensity score prediction model, regardless of function specification.⁸ The second step performed matching based on the estimated propensity scores. We used 1:1 nearest-neighbor matching such that each store i in the treatment group was matched with a store in the pool of the control group with the smallest difference in the estimated propensity score from store i . This is the most effective method for settings in which the aim is to select samples for studying the follow-up period (Stuart, 2010), which fits our research setting. Note that the purpose and advantage of using propensity score matching are to ensure that the conditional distribution of the covariates is independent of the treatment

assignment and to avoid the “curse of dimensionality” (Caliendo & Kopeinig, 2008). In this sense, the treatment group and the matched control group should have the closest propensity scores, although it is possible that a covariate could experience reduced balance after matching. Standardized bias and other balance checks are presented in the later empirical results section. We also used various other matching algorithms as robustness checks.

Our GBM specification is in Equation (1). The estimate $\hat{g}^*(x)$ adds all predicting variables' $g^*(x)$ together. $g^*(x)$ represents the sum of all regression trees that split from a predicting variable x . For example, $g_1^*(\ln shopopentime)$ refers to all regression trees that split from $\ln shopopentime$ and its following interactions with other predicting variables. $\hat{g}(X)$ is then used to model the log-odds of treatment assignment $g(X) = \log\{p(X)/[1 - p(X)]\}$ to obtain the propensity score $p(X)$.

$$\hat{g}(X) = g_0 + g_1^*(\ln shopopentime) + g_2^*(\ln credit) + g_3^*(score) + g_4^*(\ln shopcollection) + g_5^*(\ln bhalfcomment) + g_6^*(\ln monthcomment) + g_7^*(\ln weekcomment). \quad (1)$$

A set of predicting variables for the store's decision to open a Weitaio account is included in the model of Equation (1).⁹ To determine these independent variables, we scrutinized Taobao store information. We believe that the following observed variables are related to both the store's opening of a Weitaio account and the store sales (Rubin & Thomas, 1996). These variables represent different aspects of a store, including age ($\ln shopopentime$, measured by the logarithm of the store's opening date), which reflects a store's attitude toward new functions; store experience ($\ln credit$, measured by the logarithm of accumulated store credit since the store opening); store quality ($score$, which is the first component score of principle component analysis on the three highly correlated store ratings—store description rating, service rating, and logistics rating); store consumer base ($\ln shopcollection$, measured by the logarithm of the number of consumers who have bookmarked the store), which could potentially follow the store's Weitaio account; store activity level, measured by the number of reviews (take log) that a store received from its buyers about their transactions during three time periods—i.e., from the store's opening to six months before the opening of their Weitaio account ($\ln bhalfcomment$), the month prior to the opening of their Weitaio account ($\ln monthcomment$), and the week prior to the opening of their Weitaio account ($\ln weekcomment$). We found that active stores were more likely to open a Weitaio account.

⁸ We also used a probit model, and the results are qualitatively the same.

⁹ Note that these variables are from the period before a store opened a Weitaio account.

Table 1. Store-Level Summary Statistics of Content Factor Scores

Content factor	Cosmetics (N = 422)		Stationery (N = 125)		Difference t-test (Cosmetics – Stationery)
	Mean	SD	Mean	SD	
<i>informative</i>	0.116	0.122	0.101	0.116	0.016**
<i>persuasive</i>	0.007	0.044	0.007	0.034	0.000
<i>promotional</i>	0.065	0.117	0.104	0.154	-0.039***

Note: ** and *** mean that the difference between two subsamples is significant at the 5% and 1% level, respectively.

Difference-in-Differences Analysis for the Sales Effect of Firm Social Media Adoption

After PSM, we conducted a DID analysis to examine the sales effect of Weitao adoption. The analysis is at the firm level, (store) with the model specification shown below, and is estimated using ordinary least squares (OLS) regression.

$$\ln \text{halfyearsales}_{ijt} = \beta_0 + \beta_1 \text{Ifweita}_{ij} + \beta_2 \text{Aweita}_{ijt} + \beta_3 \text{Ifweita}_{ij} \times \text{Aweita}_{ijt} + \Theta \text{CV}_{ijt} + \varepsilon_{ijt} \quad (2)$$

In Equation (2), i denotes a matched pair of stores, j denotes a store from the treatment group or control group, and t denotes the time period (we use six months before and six months after the store opened a Weitao account). The dependent variable $\ln \text{halfyearsales}_{ijt}$ denotes the sales that the store received during the six months before or after the store opened a Weitao account. We think that six months is suitable for examining the firm-level Weitao effect. Later, we conducted a dynamic effect analysis from one week to one month to six months after the opening of a Weitao account. Key independent variables included Ifweita_{ij} , a dummy variable that equals 1 if store i opened a Weitao account and 0 otherwise; Aweita_{ijt} , a dummy variable that equals 1 for the period after a store opened a Weitao account and 0 for the period before; and the interaction term $\text{Ifweita}_{ij} \times \text{Aweita}_{ijt}$, which gives the impact of social media adoption on store sales performance, and its coefficient β_3 is our focus. We included a variety of control variables, CV_{ijt} , which influenced store sales: $\ln \text{shopopentime}$, $\ln \text{credit}$, score , $\ln \text{shopcollection}$, $\ln \text{halfcomment}$, $\ln \text{monthcomment}$, and $\ln \text{weekcomment}$ (as described above for Equation 1). We conducted DID analysis on both the full sample and subsamples of high- and low-involvement industries.

First Differencing Model for Sales Effect of Firm Social Media Efforts

To investigate how to manage social media efforts to achieve better firm sales performance (i.e., H3-H5), we constructed models for the examination of the number of Weitao followers, number of blog postings, and blog posting content types. These analyses were conducted on the treatment group that adopted Weitao. We also analyzed industry differences in the sales effect of social media efforts between high- and low-

involvement industries. To control for firm heterogeneity, we used a first differencing model.

Specifically, based on the main model (Equation 2), there can be equations for two time periods, i.e., the six months before Weitao account opening ($\text{Aweita} = 0$) and the six months after Weitao account opening ($\text{Aweita} = 1$). Taking the difference between the two equations, for store i in the treatment group, gives us the first differencing model:

$$\Delta \ln \text{halfyearsales}_i = \beta_2 + \beta_3 + \Theta \Delta \text{CV}_i + \Delta \varepsilon_i \quad (3)$$

In Equation (3), the dependent variable $\Delta \ln \text{halfyearsales}_i$ is the sales difference between the six months after a store opened a Weitao account and the six months before. Control variables (CVs) are the first difference of the same control variables as in Equation (1), including $\Delta \ln \text{credit}_i$, Δscore_i , $\Delta \ln \text{shopcollection}_i$, $\Delta \ln \text{halfcomment}_i$, $\Delta \ln \text{monthcomment}_i$, and $\Delta \ln \text{weekcomment}_i$, except that $\ln \text{shopopentime}_i$ was eliminated by first differencing.

Based on Equation (3), we added terms of social media efforts, i.e., the number of Weitao followers (take log) $\ln \text{follower}_i$, the number of blog postings weitaonum_i , and Weitao content types (*informative*, *persuasive*, and *promotional*), to capture their sales effect. Model specifications are shown below:

$$\Delta \ln \text{halfyearsales}_i = \alpha_0 + \alpha_1 \ln \text{follower}_i + \alpha_2 \Delta \ln \text{credit}_i + \alpha_3 \Delta \ln \text{halfcomment}_i + \alpha_4 \Delta \ln \text{monthcomment}_i + \alpha_5 \Delta \ln \text{weekcomment}_i + \alpha_6 \Delta \text{score}_i + \eta_i \quad (4)$$

$$\Delta \ln \text{halfyearsales}_i = \chi_0 + \chi_1 \text{weitaonum}_i + \chi_2 \Delta \ln \text{credit}_i + \chi_3 \Delta \ln \text{halfcomment}_i + \chi_4 \Delta \ln \text{monthcomment}_i + \chi_5 \Delta \ln \text{weekcomment}_i + \chi_6 \Delta \text{score}_i + \mu_i \quad (5)$$

$$\Delta \ln \text{halfyearsales}_i = \delta_0 + \delta_1 \text{informative}_i + \delta_2 \text{persuasive}_i + \delta_3 \text{promotional}_i + \delta_4 \text{weitaonum}_i + \delta_5 \Delta \ln \text{credit}_i + \delta_6 \Delta \ln \text{halfcomment}_i + \delta_7 \Delta \ln \text{monthcomment}_i + \delta_8 \Delta \ln \text{weekcomment}_i + \delta_9 \Delta \text{score}_i + \sigma_i \quad (6)$$

Specifically, for the sales effect of the number of Weitao followers (Equation 4), coefficient α_1 is our focus. For the sales effect of blog postings, we conducted hierarchical regression to examine the effect of the number of blog

postings (Equation 5) and the effect of blog postings' content types (Equation 6). Therefore, coefficients χ_1 and δ_1 - δ_4 are our focus.

We constructed Equations (5)-(6) separately from Equation (4) due to the different distributions of the key variables. The number of Weitao followers, Infollower_i , follows a normal distribution, and we used OLS regression in the first step of the control function method (see below for details). In contrast, for the number of blog postings, weitaonum_i , which is overdispersed (i.e., a much greater variance than its mean), we used a generalized negative binomial model.

The above models are subject to the endogeneity issue. That is, the key independent variables, i.e., the number of Weitao followers and the number of blog postings, can be influenced by store sales, i.e., the dependent variable. To address this issue, we constructed instrumental variables. Specifically, we used store i 's competitors' number of Weitao followers, i.e., Infolloweriv_i , as the IVs for store i 's number of Weitao followers and the competitors' number of blog postings, i.e., weitaonumiv_i , as the IVs for store i 's number of blog postings. The definition of "competitors" and the construction process of the two sets of IVs are as follows. First, we reviewed the search results when we searched cosmetics and stationery stores on Weitao's mobile search page for the treatment group. Stores with similar rankings in the search results are similar in many aspects, including store age, level of activity, credits, and ratings. Thus, they are competitors. Here, we define the three stores that were listed right above the focal store i in the search results and the three stores that were right below store i as "competitors." Second, we used the number of Weitao followers of these six competing stores as the IVs for store i 's number of Weitao followers. That is, there are six IVs for the endogenous variable Infollower_i . Similarly, we used the number of blog postings of these six competitors as the IVs for store i 's number of blog postings.

The instrumental variables fulfill both requirements (Wooldridge, 2002). First, the competitors' number of Weitao followers (and blog postings) are highly correlated with the focal store i 's number of Weitao followers (and blog postings) due to the similarity and competition among competitors. For our sample, the model-free evidence shows significant correlations ($p < 0.001$) between the IVs and store i 's number of Weitao followers (and blog postings). Second, when applying the six IVs individually, some competitor stores (e.g., the third one above the focal store and the third one below) are different from the focal store i ; thus, their numbers of followers and blog postings would be expected to have limited influence on the focal store i 's sales. We conducted the weak IV test and the Sargan overidentifying test for the two sets of IVs, and the results (available upon request) show that both sets are valid.

To estimate Equations (4)-(6) using these two sets of instrumental variables, we conducted the control function method (Wooldridge, 2015). For example, for the control function estimation procedure for Equation (5) to determine the number of blog postings, we first used a generalized negative binomial model to estimate Equation (7) in order to obtain $\hat{\omega}_i$ (we used OLS for the number of followers); then, we added $\hat{\omega}_i$ to Equation (5) and estimated the coefficients using OLS. The estimation procedures for Equations (4) and (6) are the same as that for Equation (5).

$$\begin{aligned} \text{weitaonum}_i = \exp(\varphi_0 + \varphi_1 \text{weitaonumiv}_i + \varphi_2 \Delta \ln \text{credit}_i + \varphi_3 \Delta \ln \text{bhalfcomment}_i \\ + \varphi_4 \Delta \ln \text{monthcomment}_i + \varphi_5 \Delta \ln \text{weekcomment}_i + \varphi_6 \Delta \text{score}_i + \omega_i). \end{aligned} \quad (7)$$

Based on the above empirical models, the summary statistics of key variables for the treatment group after opening a Weitao account are provided in Table 2.

Empirical Results

PSM Results

We first report GBM propensity score estimation results along with fit statistics. Table 3 shows the relative influence of each predicting variable. As can be seen, the store activity level (particularly $\ln \text{bhalfcomment}$ and $\ln \text{monthcomment}$) and store age ($\ln \text{shopopentime}$) are more powerful in predicting a store's decision to open a Weitao account for both the cosmetics and stationery subsamples.

We performed 1:1 nearest-neighbor matching based on estimated propensity scores to obtain the control group. After matching, we conducted a series of examinations to ensure the matching quality. First, we compared the "standardized bias" before and after matching for each predicting variable in the GBM. The standardized bias is the percentage difference of the sample means between the treatment and control groups divided by the square root of the average of sample variances of the treatment and control groups (Rosenbaum & Rubin, 1985). Compared with statistical inference such as t -tests, our method is more appropriate to measure the matching quality because it is a quantity similar to the effect size and is not influenced by sample size (Harder et al., 2010). As can be seen in Table 4, the matched groups are balanced as the standardized bias gets smaller. For matching to be acceptable, the absolute standardized bias after matching should be less than 20% (Rosenbaum & Rubin, 1985). For our sample, the bias is less than 18% for all predicting variables, which indicates that there is no significant pretreatment difference between the treatment and control groups.

Table 2. Summary Statistics of Key Variables

Variable	Cosmetics				Stationery			
	Mean	SD	Min	Max	Mean	SD	Min	Max
Inhalfyearsales ₀	5.300	2.085	0	12.937	6.385	2.410	1.609	11.986
Inhalfyearsales ₁	5.378	2.330	0	12.769	6.901	2.689	0	12.524
Inshopopentime	10.620	0.021	10.555	10.640	9.848	0.045	9.723	9.895
Incredit	6.578	2.512	2.079	15.097	7.940	2.446	3.135	13.012
score	-0.137	1.878	-31.795	1.096	-0.131	1.738	-17.083	1.474
Inshopcollection	5.242	2.233	0	13.031	6.087	2.400	1.386	11.859
Inbhalfcomment	5.843	2.744	0	14.998	7.051	2.598	2.197	12.593
Inmonthcomment	3.348	2.555	0	11.157	4.877	2.881	0	10.577
Inweekcomment	2.077	2.108	0	9.626	3.520	2.560	0	9.215
Infollower	4.703	2.271	0	12.986	5.477	2.420	0.693	11.499
weitaonum	8.611	36.128	1	498	10.762	38.708	0	315
N	422				125			

Note: (1) *Inhalfyearsales₀* is the sales in the six months before a store opened a Weitao account; (2) Please see the variable descriptions in the text below Equation (1).

Table 3. Relative Influence by GBM

Variable	Relative influence (%)	
	Cosmetics	Stationery
Inshopopentime	35.00	27.05
Incredit	3.73	5.26
score	2.57	0.15
Inshopcollection	3.65	6.70
Inbhalfcomment	50.49	19.19
Inmonthcomment	4.44	30.84
Inweekcomment	0.12	10.80
Best iteration	19,991	19,956

Table 4. Summary Statistics and Covariate Balance before and after Matching

Variable	Unmatched	Mean		Std. bias (%)	Bias reduction (%)	t-Test	
	Matched	Treatment	Control			t	p > t
Cosmetics							
Inshopopentime	U	9.853	9.811	83.7	91.5	16.04	0.000
	M	9.853	9.857	-7.1		-1.17	0.244
Incredit	U	5.965	7.600	-61.7	83.5	-12.33	0.000
	M	5.965	5.695	10.2		1.49	0.136
score	U	0.050	0.005	3.1	-362.1	0.60	0.547
	M	0.050	0.254	-14.5		-2.04	0.042
Inshopcollection	U	4.923	6.006	-46.7	71.6	-9.20	0.000
	M	4.923	4.615	13.3		2.01	0.044
Inbhalfcomment	U	3.864	6.997	-85.8	93.2	-18.83	0.000
	M	3.864	3.649	5.9		0.80	0.423
Inmonthcomment	U	3.663	3.808	-6.5	-154.5	-1.25	0.213
	M	3.663	3.293	16.7		2.61	0.009
Inweekcomment	U	2.107	2.449	-17.1	65.3	-3.27	0.001
	M	2.107	1.988	5.9		0.93	0.352
Stationery							
Inshopopentime	U	9.848	9.824	54.0	69.3	5.93	0.000
	M	9.848	9.856	-16.6		-1.43	0.153
Incredit	U	7.113	7.066	1.8	-43.8	0.20	0.838
	M	7.113	7.045	2.6		0.21	0.834
score	U	0.057	0.062	-0.5	-408.9	-0.04	0.965
	M	0.057	0.031	2.4		0.28	0.779

Inshopcollection	U	5.746	5.211	22.9	22.0	2.63	0.009
	M	5.746	5.329	17.9		1.40	0.163
Inbhalfcomment	U	5.017	6.346	-37.0	68.8	-4.52	0.000
	M	5.017	4.602	11.5		0.83	0.409
Inmonthcomment	U	4.927	3.450	61.9	92.1	6.75	0.000
	M	4.927	5.043	-4.9		-0.41	0.682
Inweekcomment	U	3.388	2.188	55.2	73.6	6.23	0.000
	M	3.388	3.704	-14.6		-1.17	0.242

Note: Bias reduction is the achieved percentage reduction in absolute bias after matching.

Second, we performed an F -test on the joint significance of all covariates and a pseudo R^2 before and after matching to measure the overall balance. Sianesi (2004) suggested reestimating the propensity score on the matched sample and comparing the pseudo R^2 with that derived before matching. After matching, there should be no systematic difference in the distribution of the covariates between the two groups. Therefore, the pseudo R^2 after matching should be fairly low. As shown in Table 5, this is true for both the cosmetics and stationery subsamples. The results of the F -tests point in the same direction, indicating a joint significance of all regressors before but not after matching. This indicates that the matched sample is much more balanced than the original sample.

Third, we conducted a stratification test, following Dehejia and Wahba (2002) to ensure that the GBM model specification is appropriate for estimating the propensity of a store to open a Weitaio account. The stratification test procedure is as follows. We divided each subsample into four intervals of the same length based on the estimated propensity scores. The interval was further split into halves if any of the covariates for the store's decision to open a Weitaio account were significantly different between the treatment and the control groups within the interval. This yielded 11 intervals for the cosmetics subsample and 8 intervals for the stationery subsample. The results (not presented due to space limits but available upon request) show that there is no significant difference between the treatment and control groups within each interval for all covariates.

Fourth, we performed a visual check by comparing the kernel density distribution of propensity scores before and after matching, as shown in Figure 1. As can be seen, there are significant differences between the treatment and control groups before matching but the differences decrease considerably after matching.

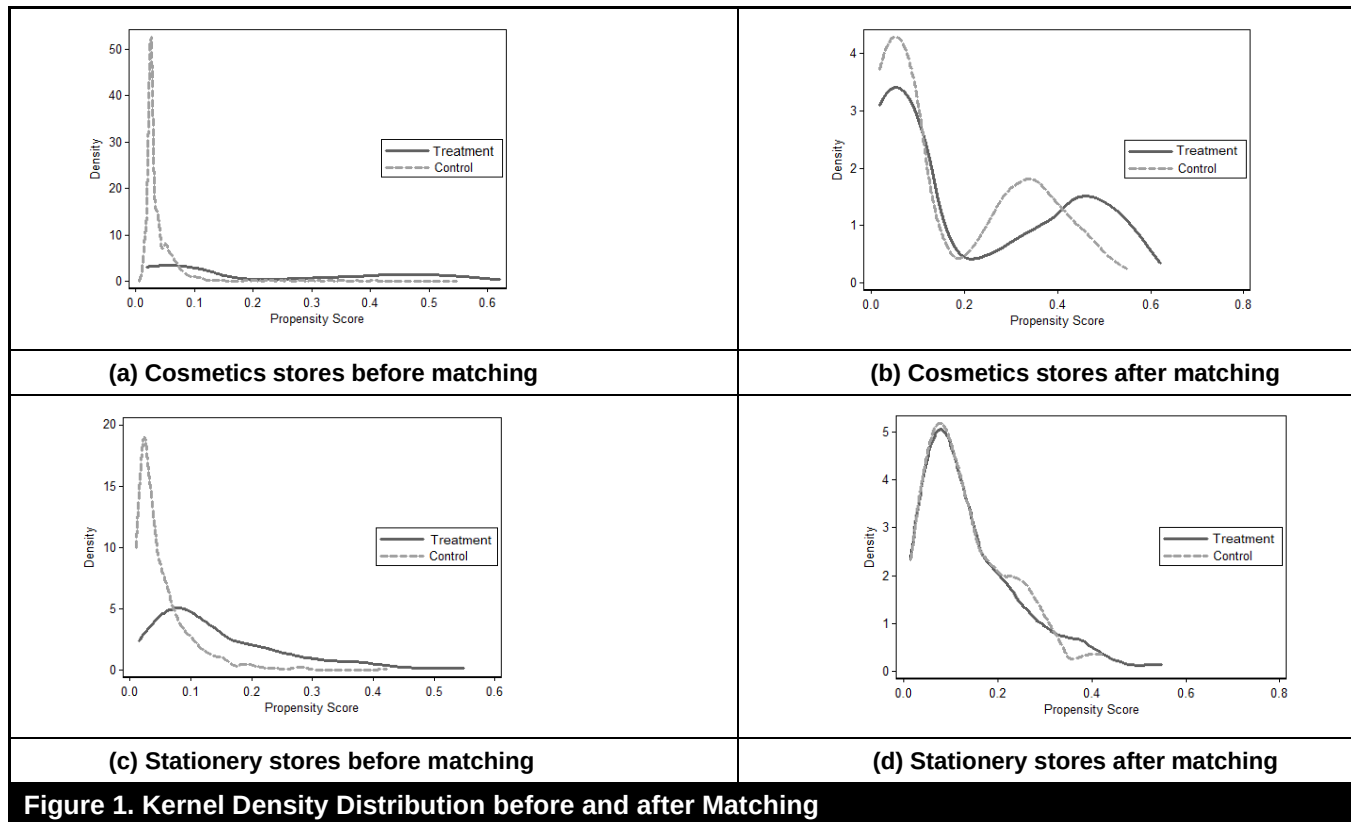
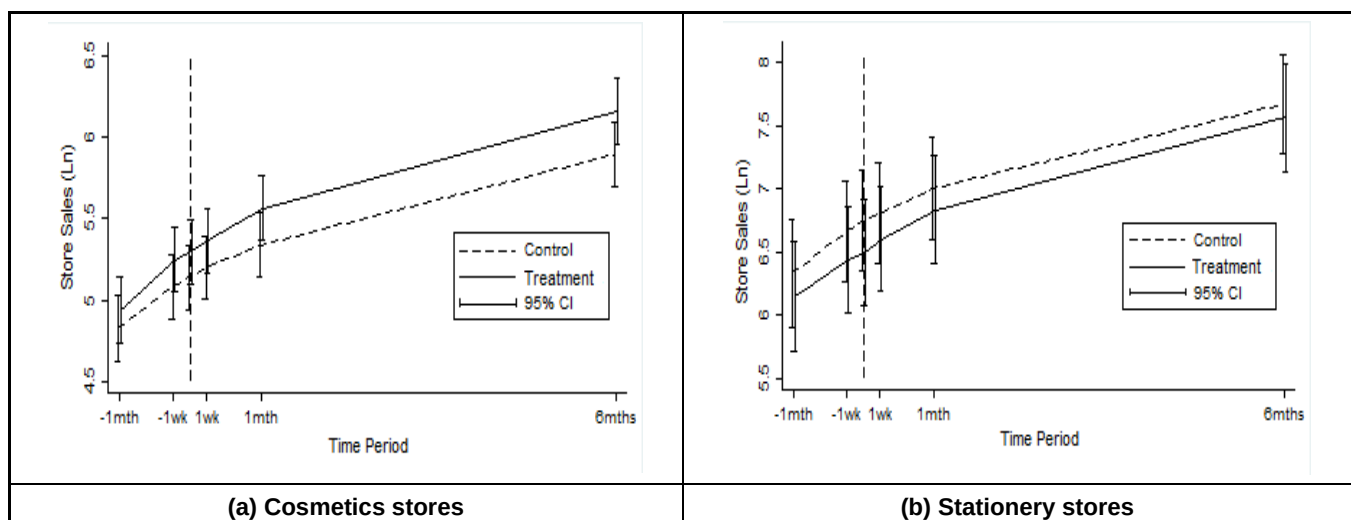
Fifth, we conducted a sensitivity analysis to assess the PSM assumption of ignorable treatment assignment, following Rosenbaum (2002) and McCaffrey et al. (2004). We used sensitivity analysis to check whether there was a hidden bias issue in the PSM, i.e., whether there were unobserved variables related to whether a store opened a Weitaio account.

The presence of a hidden bias would mean that stores with the same value as the covariate vector would have different odds of opening a Weitaio account. To conduct the sensitivity analysis, we defined the sensitivity factor (i.e., the strength of hidden bias) as Δ . $\Delta = 1$ means that a hidden bias is present at the current level; thus, larger values of Δ indicate a larger hidden bias. We let Δ increase from 1 to see how the outcome changed. If inferences about the probability of opening a Weitaio account remained unchanged as Δ increases, we can conclude that our results are robust even if it were possible to obtain and account for additional variables. According to Lin et al. (1998), if results hold when Δ is equal to or greater than 2, confidence in the results should be quite strong. Based on our sensitivity analysis, the values of Δ are 2.1 for the stationery subsample and 1.3 for the cosmetics subsample, suggesting that there is no hidden bias issue in the stationery subsample but that there may be a hidden bias problem in the cosmetics subsample. We solved this problem using the control function method (see description in the next section).

Finally, it is important to examine pretreatment differences between the treatment and control groups because the validity of the DID approach hinges on the comparability of the two groups. In addition to the standardized bias analysis and the visual check described above, we conducted a pretreatment trend analysis. We obtained observations of store sales at seven time points, i.e., accumulated store sales at six months, one month, and one week before the Weitaio account was opened; at the time the store opened the Weitaio account; and one week, one month, and six months after the Weitaio account was opened. Taking store sales at the first point as the baseline, the points in Figure 2 represent the sales trend over the following time points for the treatment and control groups; the vertical dashed line indicates the time the store opened the Weitaio account. We present 95% confidence intervals of the six time points for both groups. Along with the results of t -tests, we see in Figure 2 that for the cosmetics subsample, the difference in store sales between the treatment and control groups is insignificant before the Weitaio account openings but becomes significant after Weitaio account openings (the confidence intervals of two groups have less overlap after Weitaio account openings).

Table 5. The Pseudo R^2 and F -Test before and after Matching

	Cosmetics		Stationery	
	pseudo R^2	F -Test	pseudo R^2	F -Test
Before matching	0.172	593.78	0.151	150.71
After matching	0.023	27.81	0.067	24.43

**Figure 1. Kernel Density Distribution before and after Matching****Figure 2. Pretreatment Trend of Store Sales between Treatment and Control Groups**

For the stationery subsample, the store sales of the treatment group are lower than that of the control group before Weitaio account openings but the difference becomes smaller after the Weitaio account openings. Overall, the analysis of pretreatment differences suggests no significant differences between the treatment and control groups before Weitaio account openings but sales performance differences become apparent after stores open Weitaio accounts.

In summary, our PSM analysis involved a series of checks (including standardized bias, pseudo R^2 , F -test, stratification, visual check, sensitivity analysis, and pretreatment trend) to ensure the quality of result matching. After matching, our treatment and control groups comprised 422 cosmetics stores and 125 stationery stores, respectively, for the following DID analysis.

DID Results for Sales Effect of Firm Social Media Adoption

Our DID results for the sales effect of firm Weitaio adoption for the full sample are presented in Table 6. Model fit (R^2) is 0.941. Our focus is the coefficient of the interaction term $Ifweitaio \times Aweitaio$, which is 0.189 and significant at the 1% level, indicating a positive and significant Weitaio sales effect. The results for other variables are consistent with expectations. We computed the sales elasticity of firm social media adoption, according to the Kennedy (1981) method. The sales elasticity for the full sample is 20.67, which means that stores have a mean 20.67% sales increase after opening the Weitaio account. Overall, H1 is supported.

We then conducted the subsample analysis for the industry difference based on product involvement. To address the previously identified hidden bias issue in PSM, we used the control function method, following Wooldridge (2015). In the situation of hidden bias, a store's decision to open a Weitaio account, i.e., $Ifweitaio_i$, can be written as a function (see Equation 8) of all exogenous CVs that influence both the store's decision to open a Weitaio account and store sales. The sales that the store received before and after opening the Weitaio account can be formulated as Equation (9).

$$Ifweitaio_i = 1[\Omega CV_i + e_i \geq 0], \quad (8)$$

$$\ln halfyearsales_i = \gamma_0 + \gamma_1 Ifweitaio_i + \gamma_2 Aweitaio_i + \gamma_3 Ifweitaio_i \times Aweitaio_i + \Pi CV_i + \tau_i. \quad (9)$$

The above equations assume that e is independent of exogenous variables and follows a normal distribution with $E(\tau|e) = \rho_1 e$. τ is an unobserved factor, and an econometric problem arises because τ is not independent of $Ifweitaio_i$. To

address this problem, the idea behind the control function method is to derive a proxy variable that is conditional on the part of the endogenous variable $Ifweitaio_i$ that depends on the error term τ . If this can be done, the remaining part of $Ifweitaio_i$ will be independent of τ , and the traditional estimation will be consistent.

We used instrumental variables for the endogenous $Ifweitaio_i$ in the control function model. In the treatment group, we used store $i-1$'s competitors to instrument for store i . Store $i-1$ and store i were ranked based on the PC Taobao store search results. When we prepared the pool of control group, we searched cosmetics (or stationery) stores on PC Taobao store search page. In the search results, it had both stores that had opened a Weitaio account (i.e., Weitaio store) and stores that had not (i.e., non-Weitaio store). The Weitaio stores comprised the treatment group and their numbering i was based on the sequence in the search results. Therefore, Weitaio store i is the next treated firm after Weitaio store $i-1$ in the PC Taobao store search results. But the stores could be far from each other in the search results since there might have non-Weitaio stores listed between them. Therefore, for store i , we constructed four IVs from store $i-1$; that are the percentages of stores that had opened Weitaio among $\pm 5, \pm 10, \pm 15, \pm 20$ stores of store $i-1$ in the search results ($\pm$ means listed right below and right above), i.e., $competitors5_i$, $competitors10_i$, $competitors15_i$ and $competitors20_i$.

The four instrumental variables satisfy the IV requirements. For relevance, the percentage of stores that opened Weitaio accounts reflects a trend of Weitaio account opening in the search period. Therefore, competitors of store $i-1$ and competitors of store i had similar rates of Weitaio account openings. For the exclusion restriction, we checked the store search results and found that our treated firms were widely dispersed in the search results. That is, store $i-1$ and store i were separated in the search results with quite a few non-Weitaio stores listed between them. When ranked differently in the search results, store $i-1$'s competitors (i.e., its neighbors) were different from store i 's competitors in terms of store characteristics (such as store activity level, credit, and ratings). Therefore, store $i-1$'s competitors' social media decisions (i.e., opening a Weitaio account) would be expected to have a limited influence on store i 's sales, thus satisfying the exclusion restriction.

The detailed control function estimation procedure is provided in Appendix C. To control for intragroup correlation, we used a cluster-correlated robust estimate of variance. Our results from the cosmetics and stationery subsamples are shown in Table 7. The model fit is high for both subsamples. The coefficients of the interaction term $Ifweitaio \times Aweitaio$ are positive and significant for both the cosmetics subsample (0.194, significant at the 1% level) and the stationery subsample (0.250, significant at the 5% level).

Table 6. DID Results for Sales Effect of Weitao Adoption

Variable	Full sample
<i>Ifweitao</i>	-0.187*** (0.032)
<i>Aweitao</i>	0.019 (0.035)
<i>Ifweitao</i> × <i>Aweitao</i>	0.189*** (0.047)
<i>Inshopopentime</i>	8.510*** (0.830)
<i>Incredit</i>	0.665*** (0.027)
<i>score</i>	0.042*** (0.014)
<i>Inshopcollection</i>	-0.0003 (0.014)
<i>Inbhalfcomment</i>	-0.146*** (0.010)
<i>Inmonthcomment</i>	0.402*** (0.026)
<i>Inweekcomment</i>	0.042** (0.018)
<i>Intercept</i>	-89.940*** (8.839)
<i>N</i>	2188
<i>R</i> ²	0.941

Note: (1) ***p* < 0.05, ****p* < 0.01; (2) Clustered standard errors are in parentheses.

Table 7. Sales Effect of Weitao Adoption after Controlling for Hidden Bias

Variable	Cosmetics	Stationery	Industry difference
<i>Ifweitao</i>	0.185 (0.134)	-0.250** (0.101)	-0.036 (0.092)
<i>Aweitao</i>	0.005 (0.028)	0.029 (0.072)	0.024 (0.026)
<i>Ifweitao</i> × <i>Aweitao</i>	0.194*** (0.045)	0.250** (0.098)	0.312*** (0.067)
<i>industry</i>			-3.794*** (0.649)
<i>Ifweitao</i> × <i>Aweitao</i> × <i>industry</i>			-0.120* (0.071)
<i>Inshopopentime</i>	8.357*** (1.102)	3.911*** (1.002)	4.751*** (0.863)
<i>Incredit</i>	0.695*** (0.030)	0.550*** (0.065)	0.655*** (0.027)
<i>score</i>	0.034*** (0.009)	0.035 (0.029)	0.034*** (0.010)
<i>Inshopcollection</i>	-0.023 (0.017)	0.007 (0.033)	-0.013 (0.016)
<i>Inbhalfcomment</i>	-0.164*** (0.012)	-0.074*** (0.020)	-0.154*** (0.011)
<i>Inmonthcomment</i>	0.379*** (0.029)	0.544*** (0.084)	0.411*** (0.028)
<i>Inweekcomment</i>	0.012 (0.024)	-0.040 (0.055)	0.016 (0.022)
<i>e</i>	-0.251** (0.088)	0.026 (0.078)	-0.111** (0.060)
<i>Intercept</i>	-88.380*** (11.740)	-38.000*** (9.920)	-46.157*** (8.537)
<i>N</i>	1688	500	2188
<i>R</i> ²	0.927	0.952	0.934

Note: (1) **p* < 0.10, ***p* < 0.05, ****p* < 0.01; (2) Clustered standard errors are in parentheses.

To ensure the validity of the instrumental variables $competitors5_i$, $competitors10_i$, $competitors15_i$, and $competitors20_i$, we conducted the weak IV test. The F -statistic for the cosmetics subsample is 12.686, and that for the stationery subsample is 11.301—both are higher than 10. These results suggest that the four IVs are not weak IVs. We also conducted the Sargan-Hansen test for overidentification. The Hansen J statistic for the cosmetics subsample is 2.358 (p -value = 0.884) and 7.824 (p -value = 0.251) for the stationery subsample, implying that the IVs satisfy the required orthogonality conditions. These results suggest a valid set of instrument variables for our analysis.

We computed the sales elasticity of firm social media adoption for the two industries based on the control function estimation results. The sales elasticity for cosmetics is 21.29, meaning that cosmetics stores have a mean 21.29% sales increase after opening a Weitaobao account. The sales elasticity for stationery is higher, at 27.79, indicating a mean 27.79% sales increase after opening a Weitaobao account.

To further examine the significance of difference in the Weitaobao sales effect between the two subsamples, we used the method proposed by Wooldridge (2002). Specifically, we constructed a dummy variable, *industry*, which equals 1 for cosmetics store and 0 for stationery store. Then, we added this dummy and an interaction $Ifweitaobao \times Aweitaobao \times industry$ into Equation (2) to run a regression. The interaction term captured the difference between the two industry subsamples. We found that its coefficient is -0.120 (see last column of Table 7), significant at the 10% level, suggesting that the sales effect of Weitaobao adoption is significantly greater for a low-involvement industry than for a high-involvement industry. This is contrary to our prediction of H2; it is explained in the discussion section.

Results for Sales Effect of Firm Social Media Efforts

We examined the sales effect of firm social media efforts in the number of Weitaobao followers, the number of blog postings, and different content types. The results for the effect of the number of Weitaobao followers and blog postings on the full treated sample are presented in Table 8. For the number of Weitaobao followers, the coefficient of *lnfollower* is 0.027 (significant at 10% level), suggesting that, in general, stores with more followers have a greater sales effect. However, for the sales effect of the number of blog postings, the coefficient of *weitaonum* is insignificant. Overall, H3 is supported but H4 is not supported. Meanwhile, we compared the difference between the two industry subsamples, finding insignificant results.¹⁰

We further explored the sales effect of Weitaobao blog content. The results are shown in Table 9. For the full treated sample analysis, the coefficient of *informative* is 0.337 (significant at the 10% level); the coefficients of *persuasive* and *promotional* are insignificant. This suggests that, overall, firm informative social media content is more significant in contributing to store sales than persuasive content, thus supporting H5. We also conducted a subsample analysis and found that informative content is significant for the sales of high-involvement cosmetics stores (coefficient of *informative* is 0.508, significant at the 5% level), promotional content is significant for the sales of low-involvement stationery stores (coefficient of *promotional* is 0.685, significant at the 10% level), and persuasive content is insignificant for both subsamples.

To compare the difference between the two subsamples, following Wooldridge (2002) described above, we added the dummy variable *industry* (1 for cosmetics; 0 for stationery) and interaction terms *informative*×*industry*, *persuasive*×*industry*, and *promotional*×*industry* in the regression. The coefficient estimates are as follows: *informative*×*industry* = 1.367 (significant at the 1% level), *persuasive*×*industry* is insignificant, and *promotional*×*industry* = -0.735 (significant at the 5% level). Overall, this indicates that the effectiveness of content types is significantly different for firms selling different types of products. Specifically, informative content is more effective for the sales of selling high-involvement products (cosmetics stores), whereas promotional content is more effective for the sales of low-involvement products (stationery stores). We provide potential mechanisms to explain these results in the discussion section.

For the above analyses, we tested the validity of the two sets of instrumental variables (i.e., *lnfollower_i* and *weitaonum_i*). The corresponding coefficients of the IVs in the first stage of the control function method are all significant for both subsamples ($p < 0.01$). The strength of the IVs was assessed with a first-stage Sanderson-Windmeijer multivariate F -test of excluded instruments ($p < 0.01$), the Cragg-Donald Wald F -statistic, and the Kleibergen-Paap Wald F -statistic (both higher than the critical value of Stock-Yogo weak ID at the 10% level for both subsamples). We also conducted the Sargan-Hansen overidentifying test for the validity of the two sets of IVs. For IVs of *lnfollower*, the Hansen J statistics are 5.503 (p -value = 0.358) for the cosmetics subsample and 5.385 (p -value = 0.371) for the stationery subsample. For IVs of *weitaonum*, the Hansen J statistics are 5.888 (p -value = 0.317) for the cosmetics subsample and 5.463 (p -value = 0.362) for the stationery subsample. Based on these diagnostic statistics, we conclude that weak IV is not a concern and the two sets of instrumental variables are both valid in this study.

¹⁰ Analysis results are available upon request.

Table 8. Results for the Sales Effect of the Number of Weitao Followers and Blog Postings

Variable	Followers	Blog postings
<i>Infollower</i>	0.027* (0.016)	
<i>weitaonum</i>		0.001 (0.005)
<i>Incredit</i>	1.615*** (0.087)	1.600*** (0.089)
<i>score</i>	0.064 (0.246)	0.067*** (0.011)
<i>Inshopcollection</i>	0.013 (0.038)	0.008 (0.032)
<i>Inbhalfcomment</i>	-0.180*** (0.028)	-0.188*** (0.027)
<i>Inmonthcomment</i>	0.130*** (0.039)	0.140*** (0.038)
<i>Inweekcomment</i>	-0.103*** (0.038)	-0.108*** (0.037)
<i>w</i>	0.048** (0.026)	0.00001 (0.00003)
<i>Intercept</i>	-0.539*** (0.125)	-0.380*** (0.050)
<i>N</i>	547	547
<i>R</i> ²	0.744	0.734

Note: (1) * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$; (2) Clustered standard errors are in parentheses.

Table 9. Results for the Sales Effect of Weitao Content Types

Variable	Full sample	Cosmetics	Stationery	Industry difference
<i>informative</i>	0.337* (0.181)	0.508** (0.0229)	-0.368 (0.361)	-0.829*** (0.455)
<i>persuasive</i>	-0.362 (0.480)	-0.383 (0.345)	-0.457 (0.745)	-1.112 (1.117)
<i>promotional</i>	-0.087 (0.134)	-0.202 (0.187)	0.685* (0.359)	0.542* (0.310)
<i>industry</i>				-0.079 (0.069)
<i>informative</i> × <i>industry</i>				1.367*** (0.518)
<i>persuasive</i> × <i>industry</i>				0.678 (1.167)
<i>promotional</i> × <i>industry</i>				-0.735** (0.369)
<i>weitaonum</i>	0.001 (0.0004)	0.001 (0.0007)	0.0006 (0.0007)	0.001 (0.0004)
<i>Incredit</i>	1.625*** (0.074)	1.683*** (0.101)	1.380*** (0.085)	1.617*** (0.085)
<i>score</i>	-0.339 (0.388)	-0.619*** (0.184)	1.002** (0.477)	-0.353 (0.266)
<i>Inshopcollection</i>	0.026 (0.037)	0.031 (0.034)	-0.221 (0.173)	0.030 (0.035)
<i>Inbhalfcomment</i>	-0.205*** (0.030)	-0.212*** (0.034)	-0.133*** (0.030)	-0.204*** (0.030)
<i>Inmonthcomment</i>	0.143*** (0.034)	0.136*** (0.043)	0.149 (0.111)	0.143*** (0.030)
<i>Inweekcomment</i>	-0.113*** (0.040)	-0.105*** (0.041)	-0.102 (0.112)	-0.111*** (0.039)

<i>w</i>	0.0001 (0.0002)	0.0001 (0.00006)	-0.008 (0.007)	-7.07×10^{-7} (4.69×10^{-6})
<i>Intercept</i>	-0.389*** (0.054)	-0.433*** (0.063)	-0.276 (0.080)	-0.337 (0.066)
<i>N</i>	547	422	125	547
<i>R</i> ²	0.738	0.727	0.853	0.741

Note: (1) * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$; (2) Clustered standard errors are in parentheses.

Robustness Checks

Matching with Alternative Algorithms

We conducted various other matching algorithms of PSM to provide robustness to the 1:1 nearest-neighbor matching in the main analysis, including Mahalanobis distance matching, Euclidean distance matching, coarsened exact matching, radius matching, nearest-neighbor matching with caliper (1:3), and kernel matching. The results are largely consistent, ensuring the similarity between the treatment and control groups. The results also passed balance checks. In addition, DID results for the sales effect of Weitao adoption are positive and significant for all matching algorithms except coarsened exact matching. For coarsened exact matching (CEM), the effect of Weitao adoption is also positive but insignificant. This indicates that the sales effect of Weitao exists; however, some store information is lost due to the fact that CEM prunes both treatment and control units (Blackwell et al., 2009). Overall, our results are robust to the different matching algorithms. The results of these analyses are provided in Appendix D.

Dynamic Sales Effect of Firm Social Media Adoption

To understand how the social media sales effect is generated and sustained, we analyzed the time-varying sales impact of Weitao adoption. To analyze store sales data, we obtained seven time stamps: accumulated store sales (since store opening) at six months, one month, and one week before the Weitao account was opened; at the time the store opened the Weitao account; and one week, one month, and six months after the Weitao account was opened. These allowed us to examine the short- and long-term sales impact for one week, one month, and six months.

The seven time points constitute six time periods, i.e., three before and three after the opening of the Weitao account. Taking the third time period (i.e., one week before the Weitao account was opened) as the baseline, we specified the model as:

$$\ln sales_{it} = \alpha + \sum_{l=1(\neq 3)}^6 (Ifweitaol_i \times time_l) \beta_l + \kappa_t + \mu_i + \varepsilon_{it}, \quad (10)$$

where dependent variable $\ln sales_{it}$ denotes the sales that store i received from the first time point to the end of time period t ($t = 1, \dots, 6$), and $time_l$ is a dummy variable that indicates the l^{th} time period. Therefore, the interaction term coefficients β_4 , β_5 , and β_6 are our focus, which gives us the sales effect of the opening of the Weitao account one week, one month, and six months, respectively. κ_t is the time effect, and μ_i is the store effect. With the uneven time periods, we added a day effect instead of a time period effect to the model.¹¹

Our results of the dynamic sales effect of Weitao adoption are shown in Table 10. For the time periods before the opening of the Weitao account, the coefficients are insignificant for both subsamples, which further suggests no systematic pretreatment difference between treatment and control groups. For the dynamic effect in the cosmetics subsample, coefficients of the interaction terms are positive and significant for all periods after the opening of the Weitao account and increase over time. That is, opening a Weitao account generates an effect after adoption, and the long-term effect is larger than the short-term effect. For the stationery subsample, the coefficient of the interaction term for the sixth time period is significantly positive, which suggests that the effect of opening a Weitao account starts after a period of time. Overall, the sales effect of opening a Weitao account increases and is sustained over time.¹²

Discussion and Conclusion

Using a unique research setting, this study adds to the literature on the value of social media on firm performance. Specifically, we investigated the impact of FGC on sales performance, answering “whether,” “what,” and “how” questions.

¹¹ Due to Taobao data access restrictions, we could not obtain the time series data of store characteristics (e.g., store credit, quality score). Thus, we did not include such control variables in Equation (10).

¹² We also conducted an analysis in which we controlled for the effect of November 11 (Singles Day, a big Tmall event on the platform). The results are consistent and available upon request.

Table 10. Dynamic Sales Effect of Weitao Adoption Taking the Third Time Period as Baseline

Variable	Cosmetics	Stationery
<i>lfweitao*time1</i>	-0.047 (0.032)	0.069 (0.067)
<i>lfweitao*time2</i>	0.001 (0.006)	0.024 (0.067)
<i>lfweitao*time3</i>		
<i>lfweitao*time4</i>	0.026*** (0.005)	0.045 (0.067)
<i>lfweitao*time5</i>	0.087*** (0.023)	0.082 (0.067)
<i>lfweitao*time6</i>	0.123*** (0.032)	0.143** (0.067)
Individual and time effects included		
<i>N</i>	5064	1500
<i>R</i> ²	0.028	0.032

Notes: (1) ** $p < 0.05$, *** $p < 0.01$; (2) Standard errors are in parentheses.

The launch of the social media function (Weitao) on the Taobao e-commerce platform enabled a quasi-experiment in which we were able to trace both firm (Taobao store) social media activities and the corresponding sales impact. Using PSM and DID methods, we found that stores had an overall mean sales increase of 20.67% for a six-month time window after they adopted Weitao, thus answering the “whether” question.

We addressed the “what” question by examining firms in terms of which industries might benefit more from social media adoption, thereby contributing to the literature. However, most prior studies are based on a single industry (or single firm) or consider the overall effect without investigating industry heterogeneity. Taking product involvement as a means to differentiate industries, we found that high-involvement industry stores (cosmetics in our sample) had a relatively lower sales increase from Weitao adoption than low-involvement industry stores (stationery in our sample). This is contrary to our theoretical prediction. Due to the nature of the high-involvement industry, consumers face higher uncertainties when purchasing cosmetics products (than when buying stationery products). We found that although firm social media content aims to address uncertainties in the consumer decision-making process, for the cosmetic stores in our sample, it was insufficient to fully eliminate consumer concerns. This may indicate a learning process or an FGC strategy for firms in high-involvement industries in terms of figuring out how to exert social media efforts to create better performance. Our follow-up examination also confirms this idea.

To enrich this study by providing more practical insights, we investigated how social media efforts can yield more benefits, thus answering the “how” question. By looking at the

extensiveness, intensiveness, and content of firm social media communications with consumers, we examined the sales effect of Weitao followers, the volume of blog postings, and FGC content types, respectively. We found that firm social media efforts in extensive communications (i.e., the number of followers) led to greater sales performance. However, our findings indicate that intensive communications (i.e., the number of blog postings) do not have a significant effect—likely because of a Weitao restriction limiting the number of blog postings that a Weitao account could post per day during our study period. Meanwhile, we found no significant difference between the cosmetics and stationery subsamples for the effect of extensive and intensive communications, suggesting that attracting more followers or posting more microblogs does not differentiate between high-involvement industries and low-involvement industries in terms of generating better sales performance.

To make a difference, firms need to conduct more nuanced tactics and consider the firm or industry nature in order to address consumer purchase uncertainties. We believe that content strategies offer a solution in the social media FGC context. Indeed, our content analysis reveals that high-involvement industry stores could benefit from posting more informative content in order to eliminate consumer uncertainty; in contrast, low-involvement stores would benefit more from promotional content. We propose some potential mechanisms for these interesting findings: First, informative content is effective for selling high-involvement products where consumers often face information deficiency and greater uncertainties. Due to the nature of high-involvement industries, when purchasing products such as cosmetics, consumers need more information about product features. Therefore, firms can provide additional information about brands and products through social media channels. Tangible

and reliable information helps address consumer purchase uncertainties and is provided by informative content. Overall, we found that informative content is effective for high-involvement purchases. Second, promotional content, such as price promotions and free-shipping information, is an important FGC content type for firms in low-involvement industries. Consumers in low-involvement situations have little motivation to process information. In such situations, promotional information would thus be attractive to consumers. In fact, we found that this is a commonly used content practice in the social e-commerce setting that we studied. Third, persuasive content seems to be not as effective as the other two content types. Persuasive content tends to differentiate the focal brand or product from competitors' brands and products. However, such messages from sellers (FGC) appear to be less convincing. It is even possible that persuasive content could be interpreted as a persuasion attempt, leading to some level of distrust (Darke & Ritchie, 2007). Overall, our findings suggest that while, on average, it is relatively harder for firms in high-involvement industries to gain a sales benefit from social media adoption, they can reap higher returns with effective content strategies, particularly those that address consumer uncertainty.

The overall results were generated by a set of methods that ensure validity and rigor. In addition to the above-noted PSM-DID, we also employed the control function method to address the hidden bias issue in PSM, instrumental variables to evaluate the endogenous store decision to open a Weitao account and store social media efforts (while checking their validity), and a standard and rigorous content-coding process for text mining. We also conducted a series of balance checks to ensure PSM matching quality and used parallel assumption testing and various other matching algorithms to double-check our main results. We believe that this study provides a more complete understanding of firm social media engagement and that the empirics bolster the validity of our conclusions.

Theoretical Implications

This study contributes to the literature on the value of social media FGC and has a number of theoretical implications. First, with our overarching theoretical framework combining consumer decision process theory and social presence theory, this paper broadens the scope of consumer decision process theory to the context of firm social media management and extends social presence theory by showing how it takes effect in different stages of the consumer decision-making process. It delineates the consumer decision-making process in the context of social e-commerce and improves the understanding of the mechanisms of how FGC affects the stages of the

consumer decision-making process by creating social presence.

Second, through examining the effect of FGC on the consumer decision-making process, this study brings a theoretical perspective of product involvement to the understanding of industry differences. Indeed, product involvement is part of the consumer decision-making process and serves as an angle through which industries can be differentiated. The previous literature on this topic has not examined how product involvement theory functions in the social media context.

Third, this study enriches communication theory by investigating the content that firms communicate with consumers. We not only confirm that content is a significant aspect of communication but also identify a new type of content, promotional content, that can help FGC influence consumer purchases, especially for low-involvement firms.

Managerial Implications

Firms have devoted substantial resources to promoting their brands and products on social media. Most managers, however, are unclear about whether and how social media engagement pays off. This study provides a more complete perspective. Based on our findings, we offer the following suggestions.

First, it is beneficial for firms to embrace social media. Our results show a generally positive and significant sales impact of firm social media adoption at the firm level. Our conceptual framework and empirical methodology can help firms more accurately evaluate the performance of their social media investments, allowing them to more effectively tailor their efforts. Second, firms need to realize that the nature of their industry determines whether and how much they could benefit from social media engagement. We find that, in general, firms in low-involvement industries have a greater sales impact from using social media, but high-involvement industries can benefit from proper content strategies. This implies that firms should manage their social media differently, based on the nature of the firm. Third, having a general positive sales impact does not mean that firms need to simply open a social media community and wait for the benefits to arrive. It is important to put effort into generating optimal content and attracting consumers. Our results suggest that firm informative content is effective for increasing sales in general, and especially for selling high-involvement products; promotional content is an effective strategy for selling low-involvement products; and firms

should deemphasize the use of persuasive content. Overall, our study provides practical guidance on how to optimize firm social media efforts to improve performance.

Limitations and Future Research

This study has limitations that could be addressed in future research. First, we were only able to choose one product category each for the high- and low-involvement industries. It would be valuable to expand the data collection to other product categories and to generalize our findings of industry differences. Second, future research could explore the role of consumer characteristics and other industry or firm characteristics (including firm promotional activities) on firm decisions regarding social media adoption and on the impact of social media engagement. Doing so would generate more nuanced findings and implications. Third, while we focus on the impact of social media on sales performance, it would be interesting to examine the firm social media learning process to identify the best practices. Fourth, it would be valuable to investigate potential Weitao spillover effects, such as how Weitao product promotion might influence other products or the indirect effect from other Weitao accounts—for example, the individual accounts of key opinion leaders. Finally, future research could extend the examination to a later stage of Weitao development when the social media function had become richer to potentially reveal diverse effects.

Acknowledgments

The authors are listed alphabetically and contributed equally to this work. We would like to thank the senior editor, the associate editor and three anonymous reviewers for their valuable and constructive suggestions. We wish to acknowledge the National Natural Science Foundation of China (Grant No. 71572004, No. 71802136, No. 71729001, No. 72192845, No. 71942003), and the Shanghai Pujiang Program (Grant No. 18PJC101) for funding this research. All errors and omissions are the sole responsibility of the authors.

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Appendix A

Literature on the Value of Social Media FGC

Table A1. Literature on the Value of Social Media FGC			
<i>Impact of social media FGC on consumer behavior</i>			
Author(s) and year	Social media FGC	Consumer behavior and data source	Main findings
Erdoğan & Cicek (2012)	Survey data on firm social media marketing, including whether brand offer campaigns and relevant and popular content appear on various social media platforms	Brand loyalty of customers from survey questionnaire	Firm social media marketing positively affects brand loyalty of the customers.
Laroche et al. (2013)	Whether a consumer joins a brand community on social media	Brand trust and brand loyalty from survey data	Joining a brand community on social media has a positive effect on brand trust, which has a positive effect on brand loyalty.
Rishika et al. (2013)	Whether customers participate in a firm's social media site	Frequency of customer visits from a wine-selling specialty firm	Customer participation in the firm's social media leads to an increase in the frequency of customer visits. This effect is greater when the social media site has more postings and for customers with strong patronage, those buying premium goods, those with lower levels of buying focus, and those with deal sensitivity.
Kim & Ko (2012)	Luxury firm social media marketing activities, including entertainment, interaction, trendiness, customized, and word of mouth	Customer equity and purchase intention from survey data	Firm social media marketing activities have significantly positive effects on drivers of customer equity (including value equity, relationship equity, and brand equity) and purchase intention.
Clark & Melancon (2013)	Whether a firm uses social media for relationship management	Customer satisfaction, loyalty, and WOM from survey data	Social media investment has a positive influence on customer satisfaction, loyalty, and word of WOM mouth.
Gu & Ye (2014)	Management responses in social media	Customer satisfaction measured by customer review rating from an online travel agency	Online management responses are highly effective among low-satisfaction customers but have limited influence on other customers.
Homburg et al. (2015)	Firm active engagement in online communities	Consumer sentiment from online forums	Consumers show diminishing returns to active firm engagement, which, at high levels, can undermine consumer sentiment. This relationship holds for conversations that address consumer functional needs.
Song et al. (2019)	Movie microblogs MGC on the Weibo platform	WOM (UGC) from the Weibo platform	Volume of firm microblogs (MGC) predicts box office revenue indirectly through increased microblogs (UGC).
Meire et al. (2019)	MGC from a European soccer team's Facebook fan page	Sentiment of consumers' comments from Amazon MTurk	MGC can influence the sentiment of consumers' digital engagement (comments). Informational MGC is more influential than is emotional MGC.
Miller & Tucker (2013)	Hospitals actively managing their Facebook pages. Their postings are categorized as staff-targeted, patients-targeted, and both-targeted.	UGC, including liking, checking-in, posting, and mentioning the hospital, from Facebook	Active firm social media management drives more UGC from the organization's employees, not its clients. Untargeted content provokes employees.

Lee et al. (2018)	Firm Facebook social media marketing content, categorized as directly informative content and brand personality-related content	User engagement, including likes, comments, shares, and click-throughs	Brand personality-related content is associated with higher levels of consumer engagement. Directly informative content is associated with lower levels of consumer engagement when used alone but higher engagement levels when combined with brand personality-related content.
Colicev et al. (2018)	Valence and vividness of FGC on Facebook	Consumer marketing funnel stages, including awareness, consideration, purchase intention, and satisfaction, collected from YouGov database	FGC valence and vividness are effective for consumer funnel stages of consideration and purchase intention.
Impact of social media FGC on firm performance			
Author(s) and year	Social media FGC	Firm performance and data source	Main findings and “how”
Single industry			
Goh et al. (2013)	Marketer-generated content from an apparel firm’s fan page on Facebook	Consumer purchase expenditures, from an apparel firm	MGC has a weaker impact than UGC on consumer purchase, and its effect is only through persuasive communications, with directed content being more effective.
Chen et al. (2015)	Artists’ broadcasting activities on MySpace	Music sales from Amazon sales rank and album price/release data	Artists’ broadcasting activities have a significant effect on their music sales, and the effect comes mainly from personal messages rather than automated messages.
Xie & Lee (2015)	Social media messages on Facebook brand fan page of a fast-moving consumer goods company	Consumer offline purchase, from a fast-moving consumer goods company	Earned and owned social media have a significant and positive impact on consumers’ likelihood to purchase; however, they have no impact on the amount purchased offline with the presence of in-store promotions.
Kumar et al. (2016)	FGC (characterized by valence, receptivity, and customer susceptibility) from a wine-selling specialty firm on its Facebook fan page	Customer spending, cross-buying, and profitability, from a wine-selling specialty firm	FGC has a positive and significant effect on customer purchase behavior, and the effect of FGC receptivity is the largest compared with FGC valence and customer susceptibility.
Hewett et al. (2016)	Firm Twitter posts	Four leading U.S. banks’ quarterly levels of customer deposits, from Bloomberg	Firms benefit from using social media (i.e., volume and valence of Twitter posts) in increased deposits.
Gong et al. (2017)	Company Weibo tweets and recruited Weibo influencer retweets of company tweets	TV show viewing of a major global media company, from CSM Media Research	Company tweets directly boost show viewing, whereas influential retweets increase viewing when broadcast information is displayed.
Song et al. (2019)	Movie microblogs MGC from Weibo platform	Movie box office revenue	The volume of MGC predicts box office revenue directly and indirectly through UGC.
Multiple industries			
Colicev et al. (2016)	Brand actions on Facebook and Twitter	Brand value (published by Interbrand and Brand Finance database) for 87 brands in 17 industries (e.g., financial services, technology, automotive, luxury)	Brand actions on Facebook (number of photos posted, number of videos posted, number of status updates) have a positive effect on brand value.
Chung et al. (2020)	Firms’ communication actions (posts and responses to customer messages) on Facebook business pages	Firm market performance measured by abnormal returns and Tobin’s q for 63 firms in 13 industries (e.g., financial and insurance, information and communication, durable manufacturing)	The volume and timeliness of a firm’s responses to negative customer messages are positively and significantly associated with firm market performance.

Appendix B

Five Steps of the Content-Coding Process

Orientation Period

Based on prior studies of FGC content and ad content (e.g., Abernethy & Franke, 1996; Lee et al., 2018), we started with 20 attributes to classify informative content and persuasive content. Informative content includes all factual data on the nature of sellers' products or services, such as product brand, origins, and components, that could reduce purchase uncertainty (Abernethy & Franke, 1996). Persuasive content refers to content that convinces consumers of product benefits (Lee et al., 2018). Three research assistants who majored in information systems were hired to conduct content coding. To ensure coding quality, we explained the meaning of these 20 attributes to them and conducted two pretests.

Specifically, we collected 99 Weibao blog postings (that contained these attributes) posted by stores that opened Weibao accounts earlier than our sample stores as the pretest sample. We selected 10 to show as examples of attributes to the judges. For the remaining 89, we conducted two pretests (43 blog postings for the first one, and 46 blog postings for the second one). In the first pretest, we asked the judges to read every blog posting and to answer whether it had the 20 attributes, which was a *Yes* or *No* for each attribute. This pretest lasted about three hours, and the three judges coded independently without communication during the process. They could refer only to the 10 examples. After that, coding disagreement was discussed until a consensus was reached, and we answered questions from the judges about the coding process. Then, the judges conducted the second pretest. We calculated the Fleiss's kappa among the three judges. Fleiss's kappa is an interrater agreement measure that extends the often-used Cohen's kappa for evaluating the level of agreement among multiple judges with a binary scale. It expresses the degree to which the observed proportion of agreement among raters exceeds what would be expected if all raters made their ratings completely random. The mean of our Fleiss's kappa was 0.73, and the Fleiss's kappa value for each attribute was larger than 0.6. This indicates a substantial level of reliability (Fleiss, 1971; Landis & Koch, 1977). Given that Fleiss's kappa may return low values even when agreement is actually high, we also calculated the percentage of agreement as a robustness check. The mean percentage of agreement was 92%, which is also satisfactory.

Developing Training Set

The three judges were asked to read and judge the contents of a randomly selected sample of 300 blog postings from our final sample. Again, the judges coded the contents with no communication, but they could refer to examples we supplied from the previous stage. The 300 blog postings constitute our training set. Overall, the mean interjudge percentage agreement was 91%, and the mean value of Fleiss's kappa was 0.72, which indicates a substantial level of reliability (Fleiss, 1971; Landis & Koch, 1977).

Text Mining on Training Set

Based on the training set obtained from the previous step, we employed a support vector machine (SVM) with a radial basis function (RBF) kernel for the text classification. It was performed with the Python package *sklearn*.

To prepare the text data for classification, we first removed irrelevant information from the blog posting content, including store names and generic advertisements. Then, we conducted Chinese word segmentation for each blog posting and filtered stop words. It is worth noting that we loaded special domain lexicons in the word segmentation, including the Taobao lexicon, the cosmetics lexicon with brand names and product names, and the stationery lexicon with brand names and product names. Blog posting content after text processing was presented by vector space, and we developed word frequency matrices. The word frequency matrices were used in the SVM text classification for each attribute.

As for the text classification for a specific attribute, we set the *C* parameter as 10 and the gamma parameter as 0.001 for the RBF kernel SVM. To evaluate the accuracy of the classifier, we employed a cross-validation method with five folds. This approach uses four folds to build the model and the remaining fold to evaluate the model. This procedure is repeated five times until each of the five folds is used for validation. The above procedure was repeated once for each attribute. Thus, we repeated this process 20 times. We obtained a text classifier from this procedure to classify the final sample.

This procedure provided us with a confusion matrix for each attribute. Based on that, we calculated accuracy to evaluate the performance of the RBF kernel SVM. Overall, the mean accuracy of these 20 attributes was 84.92%, which is a satisfactory level.

Content Classification

We proceeded to the final coding process based on the SVM text classifier obtained in the previous step. Thus, we had values of the 20 attributes for each Weibo blog posting.

Factor Analysis

We then conducted factor analysis with varimax rotation to examine whether our classification of the two types of content with 20 attributes is proper. Interestingly, the results from scree plots and eigenvalues revealed three factors and 13 attributes (Table B1).

Table B1. Content-Coded Classification, Description, and Summary Statistics						
Factor	Attribute	Description	Mean	SD	Min	Max
informative	Brand	Whether it mentions a specific brand or firm name	0.151	0.358	0	1
	Product	Whether it mentions a specific product	0.388	0.487	0	1
	Packaging	Whether it mentions special packaging of the product or a special shape of the product that makes it more desirable	0.051	0.221	0	1
	Components	Whether it mentions what the product is comprised of, what ingredients it contains, and what ancillary items are included with the product	0.105	0.307	0	1
	Origins	Whether it mentions where the product is from (e.g., Korea, Japan)	0.065	0.246	0	1
	Quality	Whether it mentions product characteristics that distinguish it from other products, based on objective evaluations of product workmanship, durability, material excellence, structural superiority, personnel superiority, and special services	0.060	0.238	0	1
	Performance	Whether it mentions product-performance comparing a product with others, such as how well it performs and what it is designed for	0.059	0.236	0	1
	User experience	Whether it mentions tips or experience of product use from other users	0.038	0.191	0	1
persuasive	Safety	Whether it mentions available safety features of a particular product, comparing it with others	0.004	0.063	0	1
	Target	Whether it mentions a targeted user segment, such as women or babies	0.011	0.106	0	1
promotional	Price	Whether it mentions product cost or price and value-retention capability	0.114	0.318	0	1
	Delivery	Whether it mentions delivery information, such as delivery fee, free shipping, and discount	0.042	0.201	0	1
	Deal	Whether it mentions promotions or deals	0.054	0.226	0	1

Appendix C

Control Function Model Estimation

The estimation procedure for control function model (Equation 9) is as follows. First, we obtained:

$$\begin{aligned} E(\tau | Ifweita o, Z) &= E(E(\tau | Z, e) | Ifweita o, Z) = \rho_1 E(e | Ifweita o, Z) \\ &= \rho_1 [Ifweita o \lambda(\Omega Z) - (1 - Ifweita o) \lambda(-\Omega Z)], \end{aligned} \quad (C1)$$

where $\lambda(.) = \phi(.) / \Phi(.)$ is inverse-Mills ratio; $\phi(.)$ is probability density function of normal distribution; and $\Phi(.)$ is cumulative distribution function of normal distribution.

Then, we plugged $E(\tau | Ifweita o, Z)$ into the following equation:

$$\begin{aligned} E(\ln halfyearsales | Ifweita o, Z) &= \gamma_0 + \gamma_1 Ifweita o + \gamma_2 Aweita o + \gamma_3 Ifweita o \times Aweita o \\ &\quad + \Pi CV + E(\tau | Ifweita o, Z). \end{aligned} \quad (C2)$$

After plugging it in, we had:

$$\begin{aligned} E(\ln halfyearsales | Ifweita o, Z) &= \gamma_0 + \gamma_1 Ifweita o + \gamma_2 Aweita o + \gamma_3 Ifweita o \times Aweita o + \Pi CV \\ &\quad + \rho_1 [Ifweita o \lambda(\Omega Z) - (1 - Ifweita o) \lambda(-\Omega Z)]. \end{aligned} \quad (C3)$$

Based on the above derivation, coefficient estimates were obtained: First, we performed a probit model estimation on Equation (8) to obtain $\hat{\Omega}$ and the generalized residual $\hat{gr}_i$:

$$\hat{gr}_i = Ifweita o_i \lambda(\hat{\Omega} CV_{ii}) - (1 - Ifweita o_i) \lambda(-\hat{\Omega} CV_{ii}), \quad i = 1, 2, \dots, N. \quad (C4)$$

Second, we regressed $\ln halfyearsales_i$ on all independent variables of Equation (9) and $\hat{gr}_i$. Estimators from this regression were consistent and $\sqrt{N}$ -asymptotically normal. If coefficient γ_3 is positive and significant, we can conclude a sales effect of Weita o adoption.

Appendix D

Matching Results from Alternative Algorithms

The overall matching quality is provided in Table D1. Covariate balance check is provided in Table D2. And results of the DID analysis with different matching algorithms are provided in Table D3.

Table D1. Quality of Matching from Different Algorithms

	Mahalanobis distance matching	Euclidean distance matching	Radius matching	Nearest neighbor matching with caliper (1:3)	Kernel matching
Cosmetics					
Before matching					
pseudo R^2	0.171	0.171	0.172	0.172	0.171
F-test	574.18	573.72	593.78	593.78	574.08
After matching					
pseudo R^2	0.003	0.011	0.016	0.022	0.016
F-test	3.29	12.22	18.91	26.91	18.49
Stationery					
Before matching					
pseudo R^2	0.154	0.154	0.151	0.151	0.151
F-test	148.3	148.3	150.71	150.71	150.71
After matching					
pseudo R^2	0.009	0.032	0.031	0.042	0.027
F-test	3.1	10.67	11.04	14.80	9.80

Table D2. Summary Statistics and Covariate Balance for Different Matching Algorithms

Variable	Unmatched	Mean		Std. bias (%)	Bias reduction (%)	t-test	
	Matched	Treatment	Control			t	p > t
Mahalanobis distance matching							
Cosmetics							
Inshopopentime	U	9.854	9.811	85.8	96.7	16.96	0.000
	M	9.854	9.855	-2.8		-0.44	0.658
Incredit	U	6.024	7.602	-59.3	98.9	-11.90	0.000
	M	6.024	6.006	0.7		0.10	0.916
score	U	-0.107	0.007	-5.9	-25.9	-1.49	0.137
	M	-0.107	0.036	-7.5		-0.99	0.323
Inshopcollection	U	4.978	6.007	-44.0	96.7	-8.74	0.000
	M	4.978	4.944	1.4		0.22	0.826
Inbhalfcomment	U	3.802	6.999	-87.0	99.1	-19.18	0.000
	M	3.802	3.830	0.8		-0.10	0.916
Inmonthcomment	U	3.752	3.810	-2.6	-9.1	-0.50	0.617
	M	3.752	3.698	2.4		0.39	0.700
Inweekcomment	U	2.196	2.451	-12.6	99.9	-2.43	0.015
	M	2.196	2.196	0.0		0.00	0.999
Stationery							
Inshopopentime	U	9.848	9.823	52.1	99.2	5.59	0.000
	M	9.848	9.847	0.4		0.03	0.973
Incredit	U	7.249	7.063	7.3	94.8	0.78	0.433
	M	7.249	7.239	0.4		0.03	0.976
score	U	0.036	0.062	-0.24	-33.9	-0.22	0.828
	M	0.036	0.071	-3.2		-0.38	0.706
Inshopcollection	U	5.858	5.209	28.0	83.0	3.13	0.002

	M	5.858	5.748	4.8		0.37	0.711
Inbhalfcomment	U	5.183	6.344	-32.4	94.5	-3.87	0.000
	M	5.183	5.247	-1.8		-0.13	0.898
Inmonthcomment	U	5.051	3.447	68.1	90.5	7.19	0.000
	M	5.051	4.899	6.5		0.53	0.595
Inweekcomment	U	3.498	2.185	61.0	94.0	6.68	0.000
	M	3.498	3.420	3.7		0.29	0.775
Euclidean distance matching							
Cosmetics							
Inshopopentime	U	9.854	9.811	85.8	95.9	16.96	0.000
	M	9.854	9.852	3.5		0.52	0.600
Incredit	U	6.024	7.606	-59.5	92.0	-11.92	0.000
	M	6.024	6.150	-4.7		-0.69	0.492
score	U	-0.107	0.007	-5.9	20.2	-1.49	0.137
	M	-0.107	-0.016	-4.7		-0.59	0.553
Inshopcollection	U	4.978	6.011	-44.2	95.2	-8.77	0.000
	M	4.978	5.027	-2.1		-0.31	0.753
Inbhalfcomment	U	3.802	7.003	-87.1	84.8	-19.19	0.000
	M	3.802	4.829	-13.3		-1.76	0.078
Inmonthcomment	U	3.752	3.813	-2.8	-16.4	-0.53	0.598
	M	3.752	3.680	3.2		0.50	0.620
Inweekcomment	U	2.196	2.454	-12.8	94.5	-2.46	0.014
	M	2.196	2.210	-0.7		-0.11	0.915
Stationery							
Inshopopentime	U	9.848	9.824	52.1	98.2	5.59	0.000
	M	9.848	9.847	1.0		0.08	0.940
Incredit	U	7.249	7.063	7.3	74.1	0.78	0.433
	M	7.249	7.201	1.9		0.14	0.885
score	U	0.036	0.062	-2.4	84.9	-0.22	0.828
	M	0.036	0.031	0.4		0.04	0.968
Inshopcollection	U	5.858	5.209	28.0	57.1	3.13	0.002
	M	5.858	5.579	12.0		0.92	0.360
Inbhalfcomment	U	5.183	6.344	-32.4	69.3	-3.87	0.000
	M	5.183	5.540	-10.0		-0.71	0.480
Inmonthcomment	U	5.051	3.447	68.1	85.3	7.19	0.000
	M	5.051	4.816	10.0		0.79	0.432
Inweekcomment	U	3.498	2.185	61.0	92.5	6.68	0.000
	M	3.498	3.400	4.6		0.34	0.735
Radius matching							
Cosmetics							
Inshopopentime	U	9.854	9.811	86.2	97.8	16.80	0.000
	M	9.854	9.854	-1.9		-0.29	0.773
Incredit	U	5.979	7.560	-61.0	99.9	-12.43	0.000
	M	5.952	5.950	0.1		0.01	0.990
score	U	-0.092	0.005	-5.1	-13.9	-1.30	0.193
	M	0.036	0.147	-5.9		-1.18	0.237
Inshopcollection	U	4.929	6.006	-46.1	91	-9.30	0.000
	M	4.912	4.814	4.2		0.64	0.524
Inbhalfcomment	U	3.759	6.998	-88.3	96.5	-19.77	0.000
	M	3.819	3.706	3.1		0.42	0.674
Inmonthcomment	U	3.714	3.808	-4.2	-46.2	-0.82	0.414
	M	3.663	3.526	6.1		1.01	0.315
Inweekcomment	U	2.161	2.449	-14.3	86.6	-2.81	0.005
	M	2.111	2.150	-1.9		-0.31	0.756

Stationery							
Inshopopentime	U	9.848	9.824	54.0	69.4	5.93	0.000
	M	9.848	9.855	-16.5		-1.35	0.177
Incredit	U	7.113	7.066	1.8	-24.1	0.20	0.838
	M	7.102	7.161	-2.3		-0.17	0.864
score	U	0.057	0.062	-0.5	-855.4	-0.04	0.965
	M	0.064	0.015	4.5		0.49	0.625
Inshopcollection	U	5.576	5.211	22.9	76.1	2.63	0.009
	M	5.684	5.559	5.5		0.42	0.678
Inbhalfcomment	U	5.017	6.346	-37.0	90.5	-4.52	0.000
	M	5.128	5.002	3.5		0.25	0.804
Inmonthcomment	U	4.927	3.450	61.9	86.7	6.75	0.000
	M	4.876	5.072	-8.2		-0.64	0.521
Inweekcomment	U	3.388	2.188	55.2	70.8	6.23	0.000
	M	3.334	3.685	-16.1		-1.21	0.226
Nearest neighbor matching with caliper (1:3)							
Cosmetics							
Inshopopentime	U	9.854	9.811	86.2	85.6	16.80	0.000
	M	9.854	9.860	-12.4		-2.03	0.043
Incredit	U	5.979	7.560	-61.0	95.6	-12.43	0.000
	M	5.952	5.881	2.7		0.41	0.985
score	U	-0.092	0.005	-5.1	-60.8	-1.30	0.193
	M	0.036	0.193	-8.3		-1.77	0.077
Inshopcollection	U	4.929	6.006	-46.1	81.9	-9.30	0.000
	M	4.912	4.717	8.3		1.29	0.918
Inbhalfcomment	U	3.759	6.998	-88.3	92	-19.77	0.000
	M	3.819	3.561	7.0		0.97	0.333
Inmonthcomment	U	3.714	3.808	-4.2	-43.0	-0.82	0.414
	M	3.663	3.529	6.0		0.99	0.323
Inweekcomment	U	2.161	2.449	-14.3	82.4	-2.81	0.005
	M	2.111	2.162	-2.5		-0.41	0.682
Stationery							
Inshopopentime	U	9.848	9.824	54.0	65.6	5.93	0.000
	M	9.847	9.855	-18.6		-1.56	0.121
Incredit	U	7.113	7.066	1.8	-14.6	0.20	0.838
	M	7.074	7.128	-2.1		-0.16	0.875
score	U	0.057	0.062	-0.5	77.1	-0.04	0.965
	M	0.071	0.072	-0.1		-0.01	0.990
Inshopcollection	U	5.576	5.211	22.9	67.4	2.63	0.009
	M	5.657	5.484	7.5		0.56	0.574
Inbhalfcomment	U	5.017	6.346	-37.0	86.5	-4.52	0.000
	M	5.091	4.912	5.0		0.35	0.725
Inmonthcomment	U	4.927	3.450	61.9	81.9	6.75	0.000
	M	4.845	5.113	-11.2		-0.89	0.376
Inweekcomment	U	3.388	2.188	55.2	64.8	6.23	0.000
	M	3.305	3.727	-19.4		-1.47	0.143
Kernel matching							
Cosmetics							
Inshopopentime	U	9.854	9.811	85.7	91.6	16.46	0.000
	M	9.853	9.856	-7.2		-1.09	0.278
Incredit	U	6.024	7.602	-59.3	91.6	-11.89	0.000
	M	6.042	5.910	5.0		0.74	0.460
score	U	-0.107	0.007	-5.9	60.6	-1.49	0.137
	M	0.008	-0.037	2.3		0.39	0.698
Inshopcollection	U	4.978	6.007	-44.0	84.5	-8.74	0.000

	M	4.984	4.825	6.8		1.02	0.307
lnbhalfcomment	U	3.802	6.999	-87.0	83.5	-19.18	0.000
	M	3.957	3.428	14.4		1.92	0.056
lnmonthcomment	U	3.752	3.811	-2.6	26.8	-0.50	0.615
	M	3.717	3.674	1.9		0.31	0.757
lnweekcomment	U	2.196	2.451	-12.6	67.7	-2.44	0.015
	M	2.161	2.244	-4.1		-0.65	0.517
Stationery							
lnshopopentime	U	9.848	9.824	54.0	87.3	5.93	0.000
	M	9.848	9.850	-6.8		-0.55	0.581
lncredit	U	7.113	7.066	1.8	97.0	0.20	0.838
	M	7.099	7.1	-0.1		-0.00	0.997
score	U	0.057	0.062	-0.5	-602.0	-0.04	0.965
	M	0.063	0.027	3.3		0.34	0.737
lnshopcollection	U	5.576	5.211	22.9	56.7	2.63	0.009
	M	5.686	5.454	9.9		0.77	0.443
lnbhalfcomment	U	5.017	6.346	-37.0	99.0	-4.52	0.000
	M	5.088	5.102	-0.4		-0.03	0.978
lnmonthcomment	U	4.927	3.450	61.9	94.9	6.75	0.000
	M	4.881	4.806	3.1		0.25	0.806
lnweekcomment	U	3.388	2.188	55.2	91.3	6.23	0.000
	M	3.337	3.442	-4.8		-0.36	0.85

Note: (1) Mean: Sample mean of treatment and control groups before and after matching; (2) Std. Bias: Standardized bias in percentage; (3) Bias reduction: Achieved percentage reduction in absolute bias after matching; (4) t-test: Equality of means between the treatment and control groups; (5) Results of nearest neighbor matching with caliper 1:1 is similar to that of caliper 1:3, and are available upon request.

Table D3. DID Results for Sales Effect of Weitao Adoption Based on Different Matching Algorithms

Variable	Mahalanobis distance matching	Coarsened exact matching	Euclidean distance matching	Radius matching	Nearest neighbor matching with caliper 1:3	Kernel matching
lfweitao	-0.166*** (0.028)	-0.099** (0.039)	-0.173*** (0.029)	-0.138*** (0.025)	-0.142*** (0.028)	-0.109*** (0.026)
Aweitao	0.077*** (0.029)	0.005 (0.038)	0.094*** (0.030)	0.064*** (0.023)	0.072*** (0.028)	0.056*** (0.021)
lfweitao×Aweitao	0.114** (0.049)	0.003 (0.064)	0.110** (0.050)	0.147*** (0.047)	0.128*** (0.049)	0.127*** (0.040)
lnshopopentime	3.819*** (0.485)	6.409*** (0.814)	4.196*** (0.539)	3.960*** (0.374)	3.534*** (0.456)	4.059*** (0.346)
lncredit	0.690*** (0.031)	0.776*** (0.043)	0.703*** (0.033)	0.642*** (0.021)	0.661*** (0.024)	0.629*** (0.020)
score	0.031*** (0.010)	0.016 (0.015)	0.033*** (0.011)	0.037*** (0.009)	0.029** (0.012)	0.033*** (0.008)
lnshopcollection	-0.012 (0.014)	-0.032 (0.026)	0.003 (0.015)	-0.012 (0.011)	-0.008 (0.013)	-0.008 (0.010)
lnbhalfcomment	-0.138*** (0.011)	-0.159*** (0.015)	-0.148*** (0.012)	-0.135*** (0.009)	-0.143*** (0.009)	-0.128*** (0.009)
lnmonthcomment	0.421*** (0.031)	0.353*** (0.038)	0.422*** (0.033)	0.425*** (0.022)	0.403*** (0.026)	0.452*** (0.022)
lnweekcomment	0.004 (0.019)	0.046* (0.025)	0.001 (0.022)	0.050*** (0.015)	0.051*** (0.018)	0.024* (0.014)
Intercept	-37.317*** (4.803)	-63.112*** (8.101)	-41.141*** (5.352)	-38.561*** (3.697)	-34.389*** (4.513)	-39.562*** (3.426)
N	2052	1070	2116	20438	3290	20372
R ²	0.946	0.942	0.940	0.932	0.937	0.935

Note: (1) * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$; (2) Clustered standard errors are in parentheses.

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